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Integrated machine learning and AHP-based framework for spatial delineation of rainwater harvesting potential

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Abstract

Groundwater scarcity and its declining situation in the semi-arid area of Jaipur district, Rajasthan make it imperative to adopt scientific solutions to locate suitable rainwater harvesting sites. This study combines Geographic Information Systems (GIS), Analytical Hierarchy Process (AHP) and machine learning (ML) algorithms for delineating the RWH suitability zones. Eleven biophysical and anthropogenic factors, namely elevation (280–647 m), slope (0–35°), topographic wetness index (TWI), drainage density, soil profile, land use/land cover (LULC), rainfall (350–700 mm yr^{-1}), hillshade, aspect, lineament density, and proximity to roads, were standardized and incorporated into a spatial database. A georeferenced inventory of 150 locations was divided into training (70%; n = 105) and testing (30%; n = 45) datasets and analyzed using XGBoost, Support Vector Machine (SVM), Gradient Boosting Machine (GBM), CatBoost, and AHP models. Among the evaluated algorithms, XGBoost exhibited the highest predictive performance with a ROC-AUC of 0.78, followed by SVM (0.76), AHP (0.73), CatBoost (0.69), and GBM (0.68). The XGBoost model identified 38.1% (3902.6 km^2) of the study area as highly suitable to very highly suitable for RWH implementation, whereas 34.6% (3533.7 km^2) and 27.3% (2783.7 km^2) were classified as moderate and low-to-very low suitability classes, respectively. Variable importance analysis revealed that rainfall (15.7%), drainage density (15.6%), elevation (11.5%), and TWI (8.8%) were the dominant controls on RWH suitability. The AHP framework showed acceptable consistency (CR < 0.10) and model validation showed an overall accuracy of 88% and a Kappa coefficient of 0.64. The integrated framework successfully identified priority recharge areas in the southeastern lowlands and offers a scalable decision support tool for sustainable groundwater management and water resource planning that is resilient to climate change in semi-arid areas.

Keywords Rainwater harvesting · Machine learning · AHP · GIS · Suitability mapping · Groundwater recharge · Semi-arid regions · XGBoost

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Introduction

The world is facing a serious water shortage problem, particularly in the developing world, where urban development, economic growth and population growth will increase the strain on already limited water resources. More than 50% of the global population now lives in areas with water stress, according to recent estimates. It is estimated that this number may increase substantially to 4.8–5.7 billion people by 2050[1]. Although water accounts for about 71% of the Earth's surface, only about 2.5% is freshwater and less than 1% is readily available for human use[2]. Around one-third of the world's available freshwater comes from underground aquifers, with the agriculture, household and industrial sectors using about 42%, 36% and 27% of aquifer water, respectively[3].

Precipitation is a basic component of groundwater recharge, which enters the ground through soil and lithological substrates[4,5]. However, climate change, excessive abstraction from aquifers, unpredictable rainfall, and non-sustainable land use practices have reduced water availability significantly[6].In confronting these issues, scholars and field professionals rely more on innovative, environmentally responsible water governance methods. Rain Water Harvesting (RWH) is one of the best known and cost-effective interventions. RWH systems have been widely used in water-deficit areas due to their ability to supplement water supplies and reduce surface runoff and groundwater recharge potential[7,8]. These systems not only improve water security but also promote ecological balance and climate resilience [9].

Although Jaipur is an arid region and a semi-arid region, it shows great potential for the implementation of RWH because of its unique topography, monsoonal and increasing urban demand[10]. Water scarcity is exacerbated in this area due to geopolitical limitations, overuse of groundwater, and the development of impervious surfaces[11]. Therefore, it is crucial to identify the best locations for RWH structures for ensuring 'sustainable water resource management' in Jaipur. The constraints of effective implementation of RWH are the variability of rainfall intensity and distribution, the slope of terrain, soil permeability, reservoir capacity, vegetation cover, and the socioeconomic constraints of land ownership and technological access[12]. Hence, to optimize RWH site selection, advanced spatial planning and data-driven approaches are required[13].

In the era of Global Information Systems (GIS), Remote Sensing (RS), and Multi-Criteria Decision-Making (MCDM) methods, the analysis of complex environmental parameters and the creation of suitability maps for Rainwater Harvesting (RWH) have been revolutionized.

These tools facilitate the integration of hydrological, geomorphological, and socioeconomic datasets, enabling researchers to identify sites that are not only hydrologically viable but also economically feasible and socially acceptable[14,15].

The AHP integrated with GIS in Iran to evaluate runoff, land use, and topographic parameters for RWH site selection. Similarly[16], incorporated proximity to roads and infrastructures alongside natural features in Saudi Arabia. In India[17], demonstrated the utility of layered GIS datasets in the Deccan Plateau, while[18] validated spatial models in Sydney's peri-urban landscapes. In Bangladesh[19], demonstrated the synergy of local knowledge and GIS tools for urban water management, and[20] expanded this by analyzing spatial parameters in northwestern Bangladesh. However, very few studies have been undertaken in semi-arid Indian contexts like Jaipur, which share distinct climatic, hydrological, and socio-political characteristics.

ML-based methodologies are increasingly embedded within rainwater-harvesting planning architectures to enhance forecasting precision and overall operational system effectiveness outcomes. ML models have been applied to forecast precipitation trends, estimate potential water yields, and assist in designing adaptive harvesting systems[21–23]. By leveraging historical meteorological data and dynamic environmental variables, these algorithms enhance the operational efficiency of RWH systems and support real-time decision-making[24].

However, there is a significant gap in research work on how water scarcity and rainfall variability co-exist in the region of Jaipur and how combined ML, GIS and MCDM techniques can be applied. Although urban areas in Rajasthan have been a focus of some concern, rural and peri-urban areas are poorly represented in empirical studies, especially in topographically varied regions exposed to drought, flood, or both.

Several studies have investigated the identification and prioritization of suitable sites for rainwater harvesting (RWH) structures using remote sensing, GIS, and multi-criteria decision analysis (Haldar & Majumder, 2022; Bera & Mukhopadhyay, 2023; Jayswal et al., 2023)[25–27]. More recently, machine-learning techniques have been integrated with geospatial analysis for RWH potential mapping, with Random Forest (RF), Artificial Neural Network (ANN), Boosted Regression Trees (BRT), and other predictive algorithms demonstrating improved performance over conventional approaches (Halder & Bose, 2024; Asrade et al., 2026)[13,25]. ML-based RWH assessment has also been successfully demonstrated in hilly terrain, where RF and BRT performed better than conventional AHP-based suitability assessment (Nazir et al., 2025)[28]. Nevertheless, the application of integrated ML-based geospatial frameworks for RWH suitability assessment remains comparatively underexplored in India's hilly, topographically complex, and rainfall-constrained regions.

Thus, the primary objective of this study is to create a map that indicates the best location for the installation of rainwater harvesting structures in Jaipur, Rajasthan using an integrated machine learning and MCDM based spatial approach. To provide spatial prioritization, 11 layers will be used, including land use and land cover, soil properties, terrain slope, precipitation patterns, socio-economic metrics, and drainage-density factors.

This research seeks to propose a scientifically sound and context-sensitive solution for optimal RWH zones in Jaipur using the analytical capabilities of both ML and MCDM. The results are expected to inform local decision-makers, water resource managers, and urban planners in designing water management infrastructure that can be sustainable and resilient in the context of the Rajasthan landscape.

Material and methods

Study area

The study area is positioned in the northwestern part of India, encompassing the Jaipur district of Rajasthan, located approximately between 26°25′N to 27°10′N latitudes and 75°30′E to 76°50′E longitudes (Fig. 1). Jaipur, the capital of Rajasthan, is characterized by a semi-arid climate and forms part of the Aravalli hill ranges, interspersed with plains, undulating terrain, and isolated rocky outcrops. The effective modelling extent used in this study covers approximately 10,220 km^2, after applying the final GIS boundary mask used for suitability classification and is home to a population of over 7 million people, predominantly concentrated in urban and peri-urban zones in Census of India, 2011[29]. Jaipur's altitude spans roughly 280 to 647 m above sea level, featuring terrain composed of arid flats, rugged uplands, and intermittently dispersed plant cover overall. These landforms influence surface runoff dynamics and pose challenges to groundwater recharge due to low infiltration capacity in many parts. The district receives a mean annual rainfall of approximately 650 mm, although considerable spatial variability exists across Jaipur, with annual precipitation ranging from about 350 mm in the drier western sectors to nearly 700 mm in the southeastern and elevated regions (IMD, 2021). However, intra-annual variability and erratic rainfall distribution often lead to water scarcity during dry months.

The average temperature varies considerably from one season to another in Jaipur. The summer temperature is often above 40 °C, the winter temperature is around 5 °C, and the average seasonal variation ranges from 24 °C to 44 °C during summer and 5 °C to 26 °C during winter. Over-pumping in the region has resulted in a steady lowering of the groundwater table, which has been compounded by poor natural percolation on rocky and compact soils. In the semi-arid region, the erratic rainfall pattern and growing demand for water necessitate the need for alternative water conservation methods such as RWH. The topographical conditions such as rocky hills and slopish ground in parts of the Aravalli foothills, coupled with urbanization, emphasize the importance of spatial planning tools to optimize the placement of RWH infrastructure.

Therefore, the present study is focused on the identification of suitable zones for RWH in Jaipur by applying a combination of geospatial and machine learning methods. Thematic layers, including land use/land cover, slope,

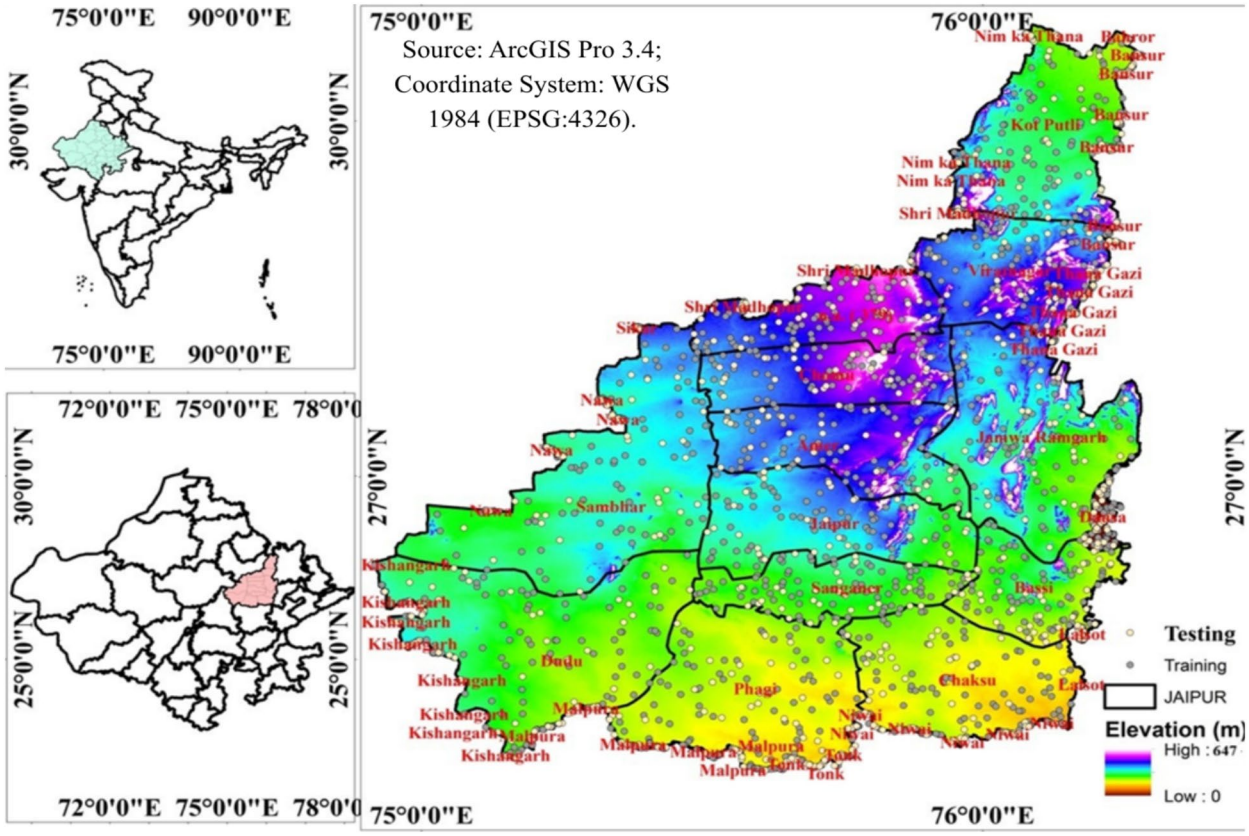


Fig. 1 Investigative zone of the study area (Jaipur District).

soil profile, rainfall, and socio-economic variables, are combined to create a suitability map for use in future water resource planning and sustainable urban development, as shown in Fig. 1.

This map is showing the zone of investigation within the district of Jaipur state of Rajasthan, India (Fig. 1). Supplemental panels depict the spatial position of the study region across the national and state scales as well. The distribution of training points (black) and testing points (yellow) used in RWH suitability modeling have been plotted. The elevation is displayed in meters above mean sea level, with a gradient of colors used to show the topographical differences throughout the area.

Inventory data

A spatial database was created using an inventory approach that combines field observations, high-resolution satellite data, topographic data, and expert knowledge of local hydrological conditions to delineate potential rainwater harvesting (RWH) zones in the Jaipur district, Rajasthan. A total of 150 georeferenced reference locations were compiled and categorized as suitable and non-suitable for RWH implementation. The classification framework used multiple environmental factors, such as surface characteristics, slope conditions, infiltration capacity, proximity to drainage networks, runoff accumulation potential and feasibility of RWH structures[30–32].

Sites located in areas with low elevation gradient, permeability, moderate to high drainage density, and favorable runoff accumulation conditions were considered as suitable sites and assigned class value = 1[33]. On the other hand, areas with steep topography, rocky ground, highly urbanized land cover, low infiltration capacity, and insufficient runoff retention were classified as nonsuitable sites (class value = 0)[34] . To keep away from class imbalance, classification bias, and equivalent illustration of both classes throughout model coaching and validation, a balanced inventory of 75 suitable and 75 non-suitable websites was intentionally used.

While detailed field photographs and site sheets could not be obtained for all reference locations, the inventory database was verified through external cross-referencing with high-resolution Google Earth imagery, terrain information derived from the DEM, topographic maps and expert interpretation to minimize classification uncertainty and enhance the reliability of the reference inventory. The resulting inventory reflects the diversity of geomorphological and hydrological conditions that are present throughout Jaipur district, and can serve as representative samples for supervised machine learning modelling[35]. The inventory dataset was randomly divided into a training set (70%, n = 105) and a testing set (30%, n = 45). The training set was used to calibrate the model, optimize hyperparameters, and for cross-validation, while the independent testing set was left for the final evaluation of the model performance assessment. To prevent overfitting and enhance the model's ability to generalize, only a ten-fold cross-validation procedure was applied to the training set[36].

Under this framework, the training dataset was partitioned into ten approximately equal folds to implement a 10-fold cross-validation strategy. During each iteration, nine folds were used for model training, while the remaining fold served as the validation set. This procedure was repeated until each fold had been used once for validation, thereby reducing the influence of data partitioning and providing a more robust estimate of model generalization. Hyperparameter optimization for all machine-learning algorithms was performed using grid search integrated with 10-fold cross-validation. Multiple combinations of model-specific hyperparameters were systematically evaluated to identify the configuration that provided the best balance between predictive performance and overfitting control. The optimal hyperparameter combination for each model was selected primarily by maximizing the area under the receiver operating characteristic curve (ROC-AUC) across the validation folds. The investigated search ranges, final optimized hyperparameters, and computational settings for each algorithm are provided in Supplementary Tables S3 and S4.

Spatially explicit validation approaches, such as spatial block cross-validation, can reduce the potential influence of spatial autocorrelation and generally provide more conservative estimates of predictive performance. However, owing to the limited number and spatial distribution of available reference observations, the present study adopted a standard random sampling strategy for model training and validation. Consequently, some degree of spatial dependence among neighbouring observations cannot be completely excluded. The reported model performance metrics should therefore be interpreted with consideration of this potential spatial autocorrelation and the possibility of slightly optimistic estimates of predictive accuracy.

To improve the robustness and representativeness of the modelling framework, a balanced inventory-based approach was adopted, incorporating both suitable and unsuitable locations distributed across the major geomorphological, topographic, and hydrological conditions of Jaipur district. This sampling strategy ensured that the training dataset captured the environmental variability associated with rainwater-harvesting suitability while minimizing potential class imbalance. The resulting inventory was subsequently used to train supervised machine-learning algorithms, including Extreme Gradient Boosting (XGBoost), Support Vector Machine (SVM), Gradient Boosting Machine (GBM), and CatBoost. These algorithms were selected because of their ability to model complex, nonlinear relationships and interactions among multiple environmental predictors, enabling spatially explicit prediction of RWH suitability across the study area.

Influencing factors of rainwater harvesting (RWH)

In this current study, eleven critical biophysical and anthropogenic factors have been considered to analyze spatial suitability for RWH in the Jaipur district of Rajasthan. The factors selected are those that have the most direct influence on the dynamics of surface runoff, on infiltration potential, and on the feasibility of water storage in conditions of semi-arid climate[37,38].The variables that are integrated are:

1. Aspect—Directional orientation of slope affects solar radiation, rate of evapotranspiration, and soil water retention[38].

2. Distance from Road Network – affects construction feasibility and access to RWH infrastructure, where a closer distance is associated with greater suitability for RWH implementation[39].
3. Drainage density—is the ratio of total stream length to watershed area, and it is a hydrological indicator of surface runoff and infiltration potential[40].
4. Elevation—affects gravitational flow, water movement, and catchment delineation, higher elevations may lessen accumulation of runoff[41].
5. Hillshade—a terrain-derived factor indicating illumination of the sun; it is related to surface evaporation and microclimatic conditions[42].
6. Lineament Density—Reflects fracture/fault planes that promote water percolation into the subsurface and is an important factor in groundwater recharge[43].
7. Land Use/Land Cover (LULC) – controls runoff generation and infiltration variability; built-up areas have lower infiltration, and forest and agriculture have higher infiltration[44] .
8. Slope – impacts the velocity of runoff and infiltration, and a gentler slope may be better for RWH structures because the movement of surface water is slower[45].
9. Topographic Wetness Index (TWI) – a hydrological index that integrates slope and upstream contributing area, and determines the zones of moisture accumulation[46]
10. Rainfall–The precipitation, its spatial distribution and quantity, directly affect the RWH potential, as indicated in the primary input variable[47].
11. Soil Profile—affects infiltration capacity and subsurface water retention depending on rock type, porosity, and permeability[48].

Geospatial and remote sensing data were employed in a GIS-based environment to process these parameters. Thematic categorization of all layers was used for resampling, normalization, and classification prior to input into the 'multi-criteria decision-making model' (MCDM) for composite suitability analysis. The spatial distribution and criteria for each influencing factor are detailed in Table 1 and an extensive methodological framework is presented in Fig. 2. Taking into account various environmental and anthropogenic factors, this approach offers a rich assessment of Jaipur's semi-arid rainwater-harvesting potential, where water resources are occasionally limited and precipitation patterns are variable. This scientific framework is designed to help planners and water managers determine areas where the use of RWHs could have maximum hydrological and socio-economic benefits.

Rainwater harvesting (RWH) potential within the study area has been spatially modelled using eleven key biophysical and topographic parameters known to influence surface water accumulation and runoff dynamics. These parameters include: Elevation (m), which reflects terrain height and affects gravitational water flow; Topographic Wetness Index (TWI), indicating moisture accumulation potential; Slope (°), representing the steepness of terrain; Aspect (°), denoting the directional orientation of slopes; Drainage Density, which influences surface runoff and infiltration; Distance from Road (km), accounting for anthropogenic accessibility and impact; Rainfall (mm), a critical climatic input for water availability; Soil Profiles, affecting infiltration and water retention capacity; Hillshade (°), derived from elevation data to assess surface illumination and micro-topographic effects; Lineament Density, indicating structural controls on groundwater movement; and (LULC, reflecting surface characteristics that modulate runoff, infiltration, and evapotranspiration.

Thematic parameters for RWH suitability in Jaipur, Rajasthan

Elevation

Elevation critically influences hydrological processes, particularly rainfall distribution, surface runoff patterns, and potential water retention zones.In the study area of Jaipur, elevation ranges from approximately 280 to 647 m above sea level, reflecting significant topographic heterogeneity across the Aravalli foothills and plains. Higher elevation zones are often associated with increased Topographic precipitation, while lower-lying areas

Table 1 Description of the key spatial and thematic parameters employed in the study.

Sl. No	Thematic layer	Data source	Spatial resolution	Remarks
1	Digital elevation model (DEM)	USGS earth explorer (1 arc-second SRTM), WGS 1984, UTM Zone 43N	30 m	Used for generating slope, aspect, TWI, and hillshade layers
2	Land use / land cover (LULC)	Landsat 8 OLI/TIRS satellite imagery via USGS Earth Explorer	30 m	Classified to assess surface characteristics and runoff potential
3	Hillshade	Derived from DEM using ArcGIS Spatial Analyst tools	30 m	Terrain illumination index derived from DEM-based simulation of sun azimuth and elevation
4	Slope	Computed from DEM using slope algorithm in GIS	30 m	Influences water runoff velocity and infiltration capacity
5	Drainage density	Extracted from DEM using hydrological tools in GIS	30 m	Indicator of surface water accumulation and runoff concentration
6	Topographic Wetness Index (TWI)	Calculated from DEM using standard GIS hydrology tools	30 m	Identifies areas with higher potential for moisture and water accumulation
7	Soil profile	FAO/UNESCO Global Soil Map Database (FAO GeoNetwork)	~30 m (resampled)	Provides information on soil permeability and water retention capacity
8	Rainfall	Indian Meteorological Department (IMD), gridded data	30 m	Average annual rainfall used to estimate water availability
9	Aspect	Derived from DEM using terrain analysis tools	30 m	Determines slope direction affecting solar exposure and evaporation
10	Distance from roads	Computed as Euclidean distance using ArcGIS 10.8	30 m	Evaluates accessibility and suitability for construction of RWH structures
11	Geological characteristics	Geological Survey of India (GSI) open datasets and maps	Vector-based (converted to raster at 30 m)	Determines rock permeability and recharge potential

Note: Hillshade was incorporated as a secondary topographic indicator to represent terrain illumination and shading effects rather than direct solar radiation measurements.

offer flatter surfaces conducive to the installation of storage structures[49]. These geomorphic characteristics play a pivotal role in determining feasible locations for RWH infrastructure as shown in Fig. 3.

Topographic wetness index (TWI)

The concept termed 'Topographic Wetness Index' (TWI) represents how soil water distribution varies, relying on the extent of contributing terrain and gradient; its computation employs the expression as Eq. 1.

$$\mathrm{TWI} = \ln\left(\frac{A}{\tan\beta}\right) \tag{1}$$

where:

- A = Upslope contributing area.
- β = Local slope (in radians).

Across Jaipur, terrain irregularities influence TWI readings, where elevated metrics clearly reveal zones possessing increased capacity for moisture collection and below-ground hydrologic percolation. Such zones are considered suitable for RWH owing to their ability to retain and percolate water effectively[50] as depicted in Fig. 4.

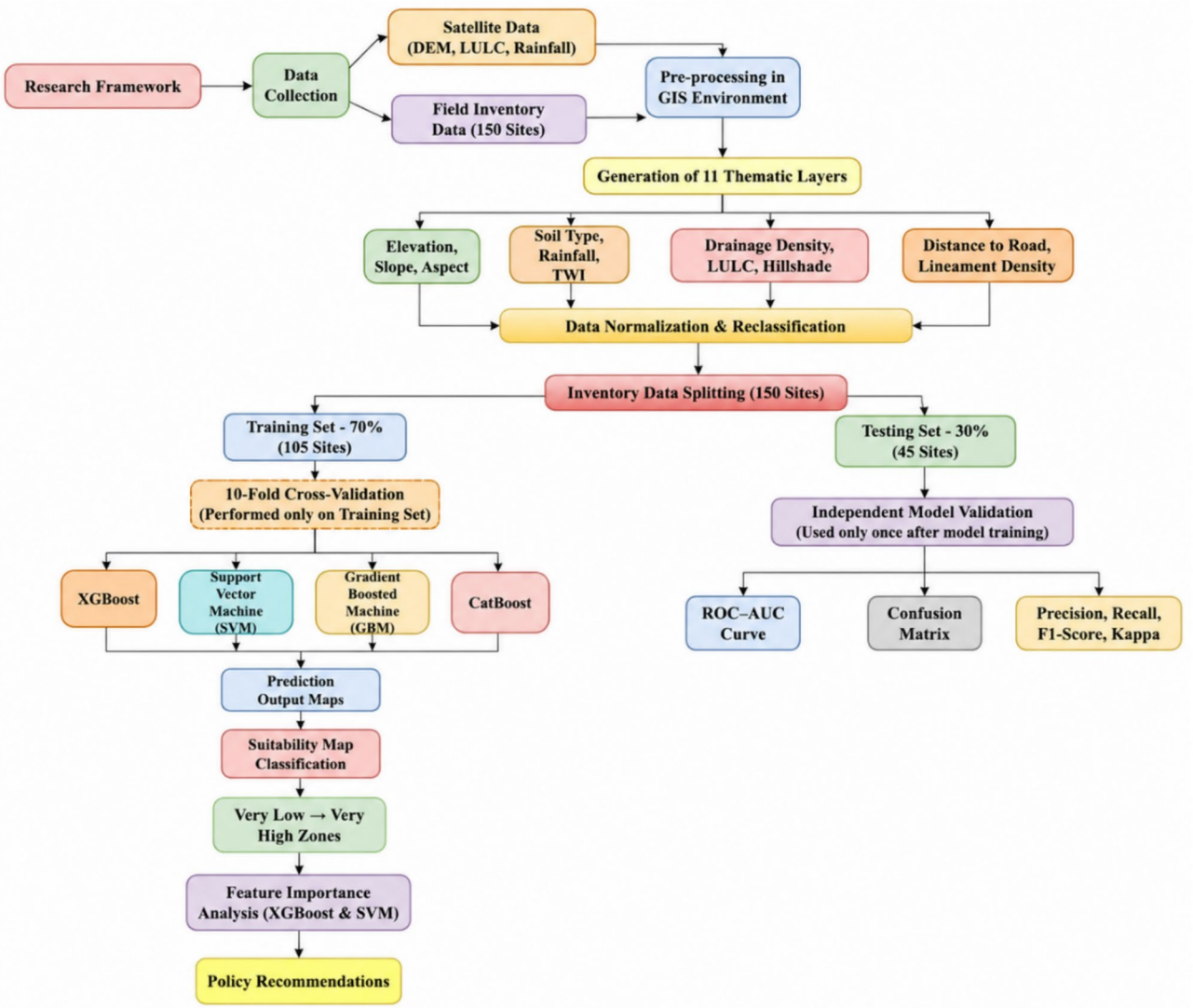


Fig. 2 Methodology flowchart.

Slope

Terrain inclination governs how fast and where runoff moves, thereby shaping its efficiency of water collection. In Jaipur, slopes range from nearly flat (0°) to steep gradients (up to ~ 35°) in hilly zones. Flatter slopes facilitate longer water residence times, encouraging infiltration and storage, whereas steeper slopes promote runoff and erosion, reducing RWH feasibility[51]. Thus, gentle slopes are prioritized for establishing RWH tanks and recharge pits as illustrated in Fig. 5.

Aspect

Aspect, denoting the compass orientation of terrain slopes, influences solar radiation exposure, evaporation rates, and microclimatic conditions. In arid environments such as Jaipur, north- and northeast-facing slopes typically experience lower evapotranspiration, offering thermally favorable conditions for minimizing water loss from

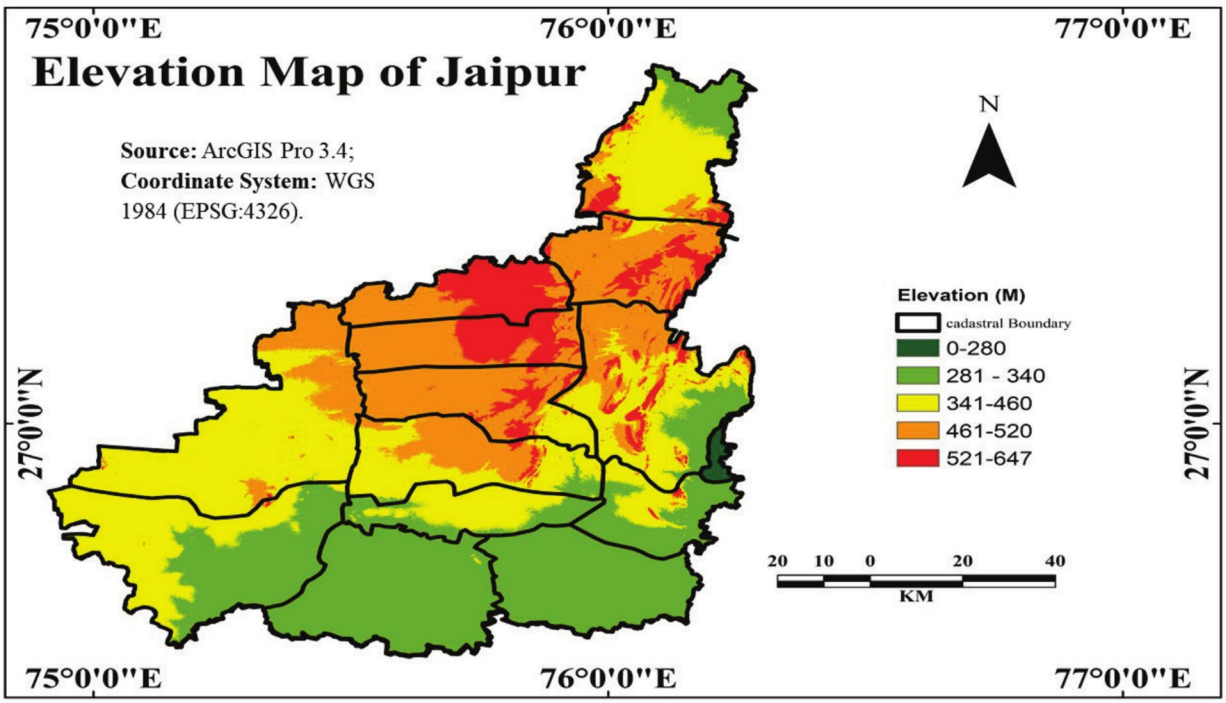


Fig. 3 Elevation map for identifying suitable rainwater harvesting (RWH) zones in Jaipur District, Rajasthan.

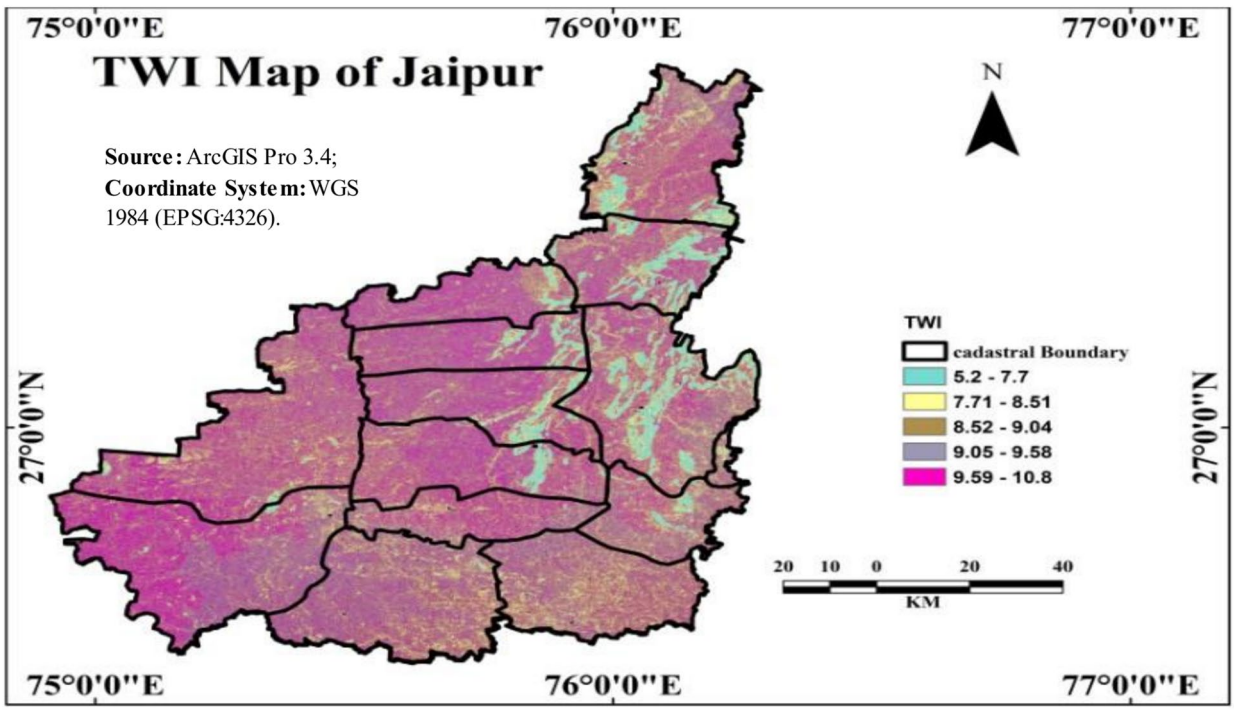


Fig. 4 TWI map to identify suitable RWH zones in Jaipur District, Rajasthan.

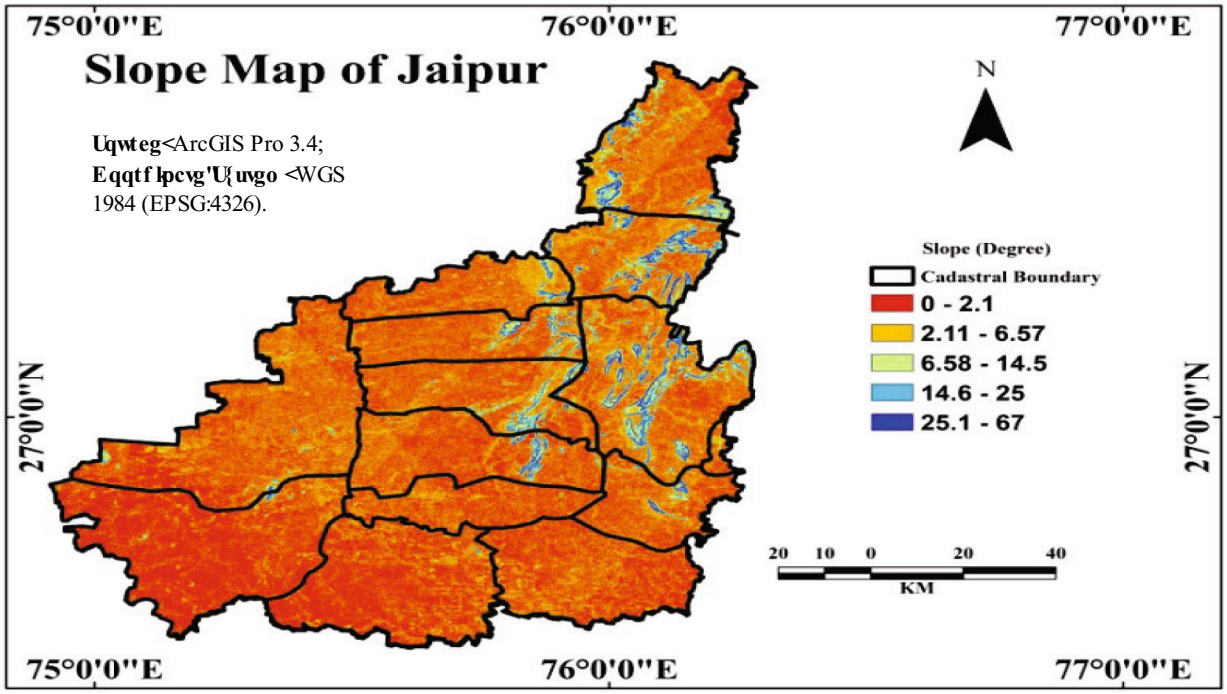


Fig. 5 Slope map to identify suitable RWH zones in Jaipur District, Rajasthan.

storage units. South-facing slopes, conversely, are subject to higher thermal load and may negatively affect the quality and volume of harvested rainwater[51] as depicted in Fig. 6.

Drainage density

Drainage network density, defined by cumulative channel distances within a given area, indicates how effectively overland flow becomes spatially distributed. Lower drainage density zones in Jaipur, particularly in the western outskirts, exhibit higher infiltration potential and are more favourable for water harvesting. In contrast, high drainage density regions in the hilly northeast may facilitate rapid runoff, limiting water retention[52] as given in Fig. 7.

Distance from road

Proximity to road networks affects the accessibility and logistical feasibility of RWH system implementation. Areas closer to roads (0–5 km) are generally preferred due to ease of transportation of construction materials and reduced maintenance costs[53]. In Jaipur, this parameter supports infrastructure planning in semi-urban peripheries where land is both accessible and cost-effective as shown in Fig. 8.

Rainfall

Rainfall is the most critical climatic variable affecting RWH. Jaipur receives yearly rainfall between 350 and 700 mm, primarily during the southwest monsoon (June–September). Spatial variation in rainfall was observed using IMD data, and higher precipitation zones in the eastern region of Jaipur district were considered optimal for harvesting (IMD, 2023)[54]. Temporal and spatial rainfall trends have been incorporated using rainfall isohyets and gridded datasets as shown in Fig. 9.

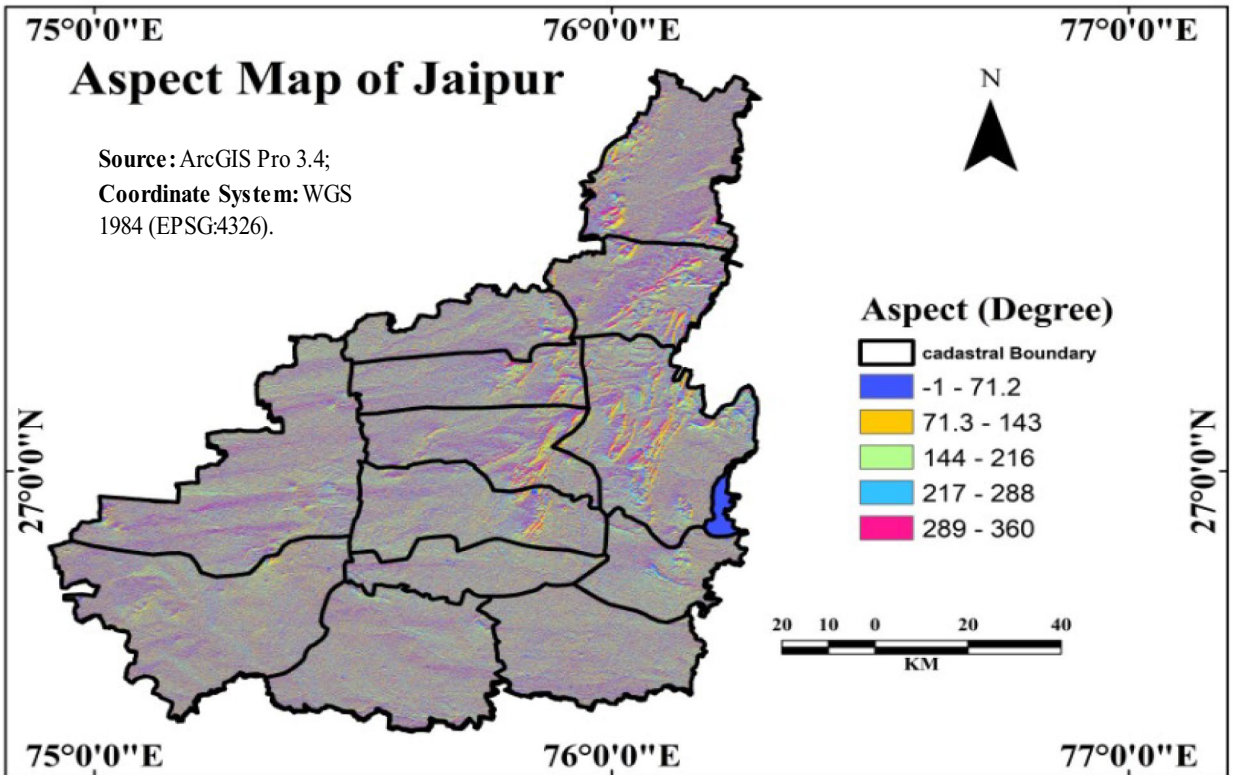


Fig. 6 Aspect map used to identify suitable RWH zones in Jaipur District, Rajasthan.

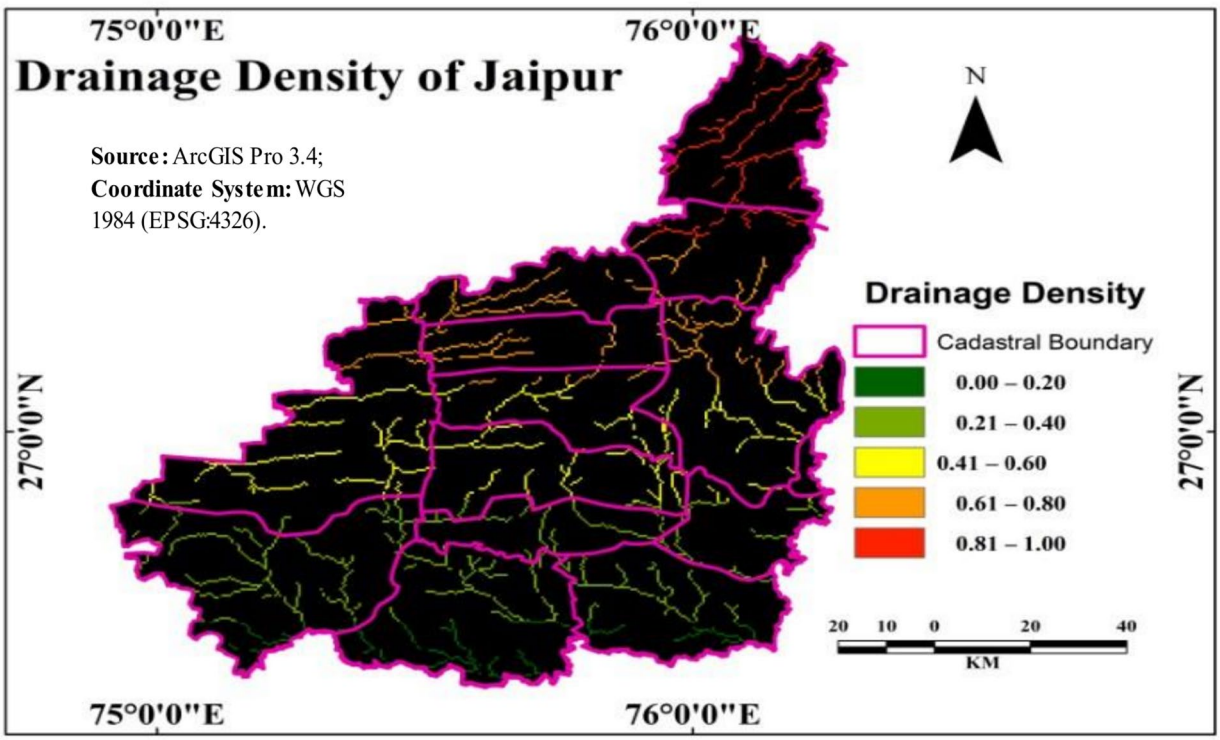


Fig. 7 Drainage density map used to identify suitable RWH zones in Jaipur District, Rajasthan.

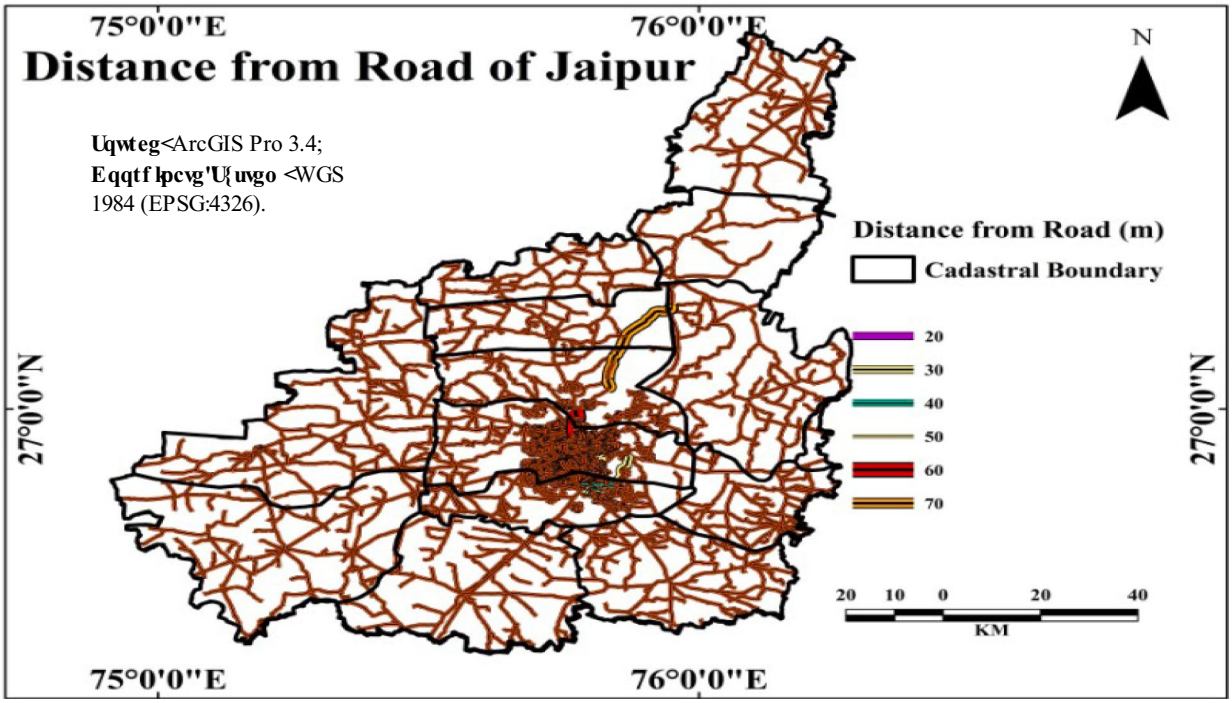


Fig. 8 Distance from road map identify RWH zones in Jaipur District, Rajasthan.

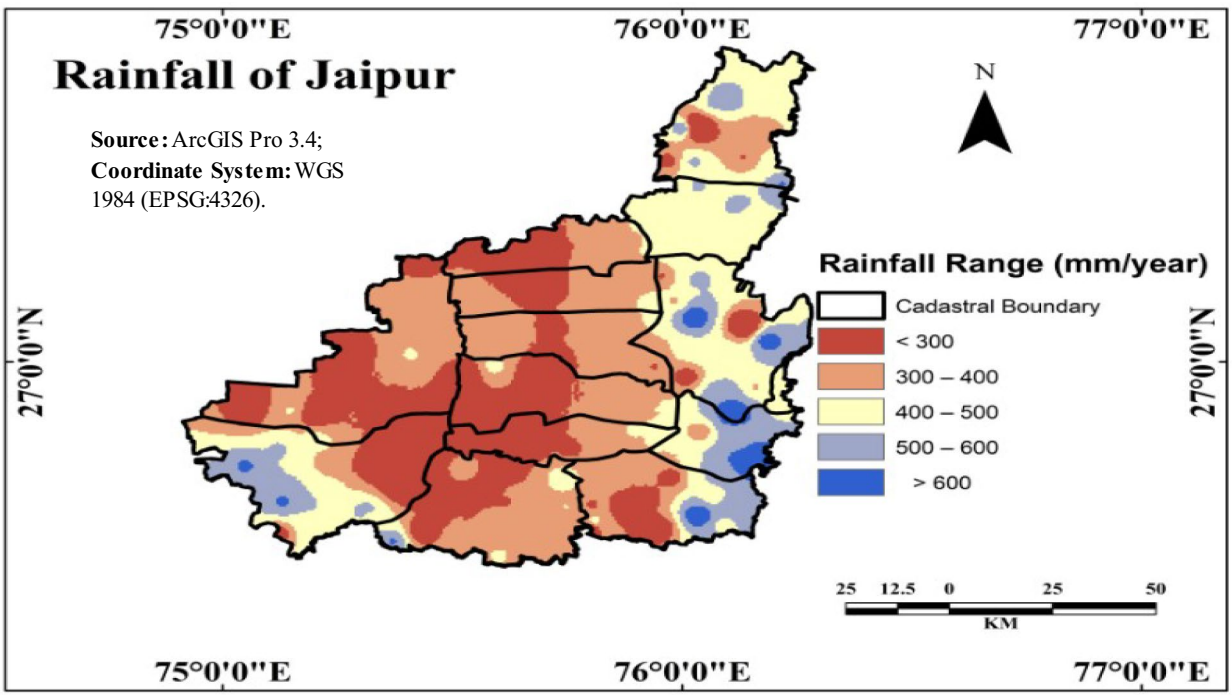


Fig. 9 Rainfall map to identify suitable RWH zones in Jaipur District, Rajasthan.

Soil

Soil texture and composition govern infiltration capacity and water-holding potential. The soils of Jaipur district exhibit considerable spatial variability and are generally characterized by coarse- to medium-textured soil groups, including sandy and loamy soils. The soil information used in this study was derived from the FAO/UNESCO Global Soil Map and was employed primarily to represent regional soil characteristics relevant to rainwater harvesting suitability assessment[55]. Sandy soils exhibit higher permeability but limited retention, whereas clay-rich soils retain more water but offer reduced infiltration (FAO, 2020)[56,57]. Suitability mapping considered these variabilities by integrating soil hydraulic conductivity and porosity indices as shown in Fig. 10. The FAO/UNESCO Global Soil Map provides generalized soil information at a regional scale and was therefore used to characterize the broad spatial distribution of soil properties rather than to infer detailed local soil textures. Consequently, the soil classes considered in this study should be interpreted as indicative representations of regional soil conditions[58].

Hillshade

Hillshade is a terrain-illumination index derived from a digital elevation model (DEM) that simulates the relative brightness of the land surface based on specified solar azimuth and elevation angles. In this study, hillshade was generated from the SRTM DEM using a solar azimuth of 315° and a solar elevation angle of 45°. Although hillshade is not a direct measure of solar radiation, it serves as a proxy for terrain illumination and topographic shading, which can influence soil moisture retention, evapotranspiration patterns, and runoff generation, thereby affecting the spatial suitability of rainwater harvesting.. Derived from Digital Elevation Models (DEM), hillshade values range from 0 (complete shadow) to 254 (full illumination). Shaded areas, particularly on leeward slopes, tend to have reduced evaporation rates, enhancing stored water longevity[59]. In Jaipur, hillshade analysis informed site selection by highlighting microclimates suitable for storage as shown in Fig. 11.

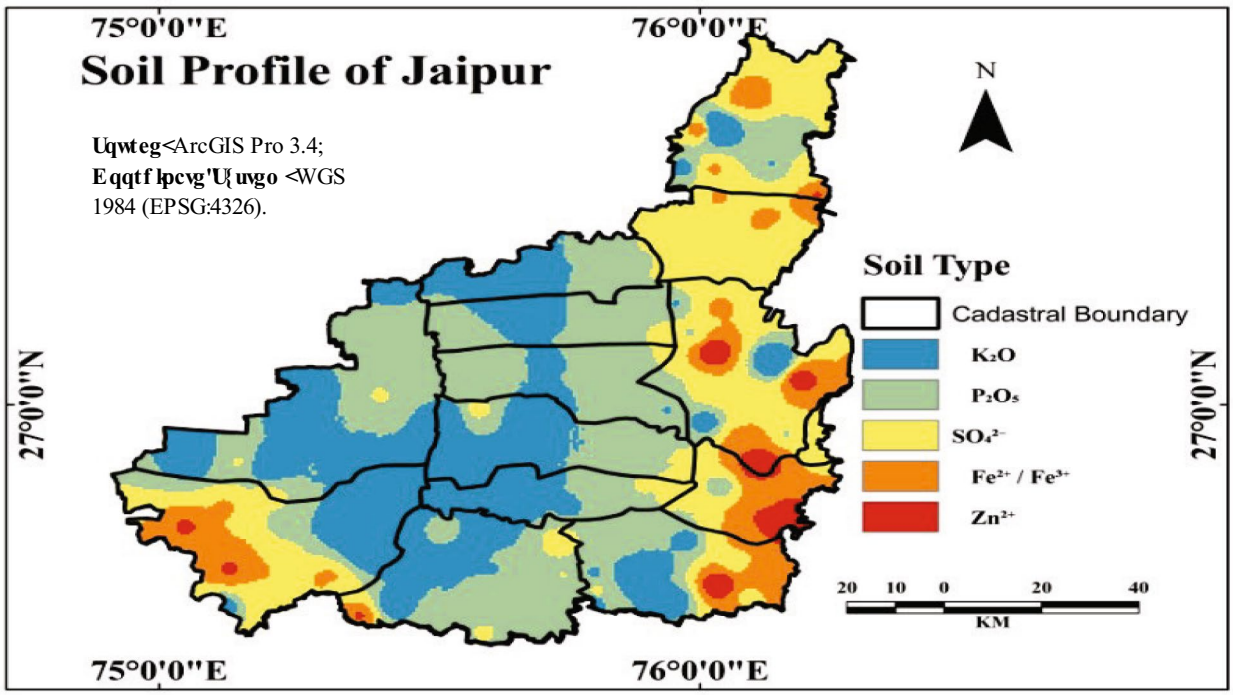


Fig. 10 Soil type map used to identify suitable RWH zones in Jaipur District, Rajasthan.

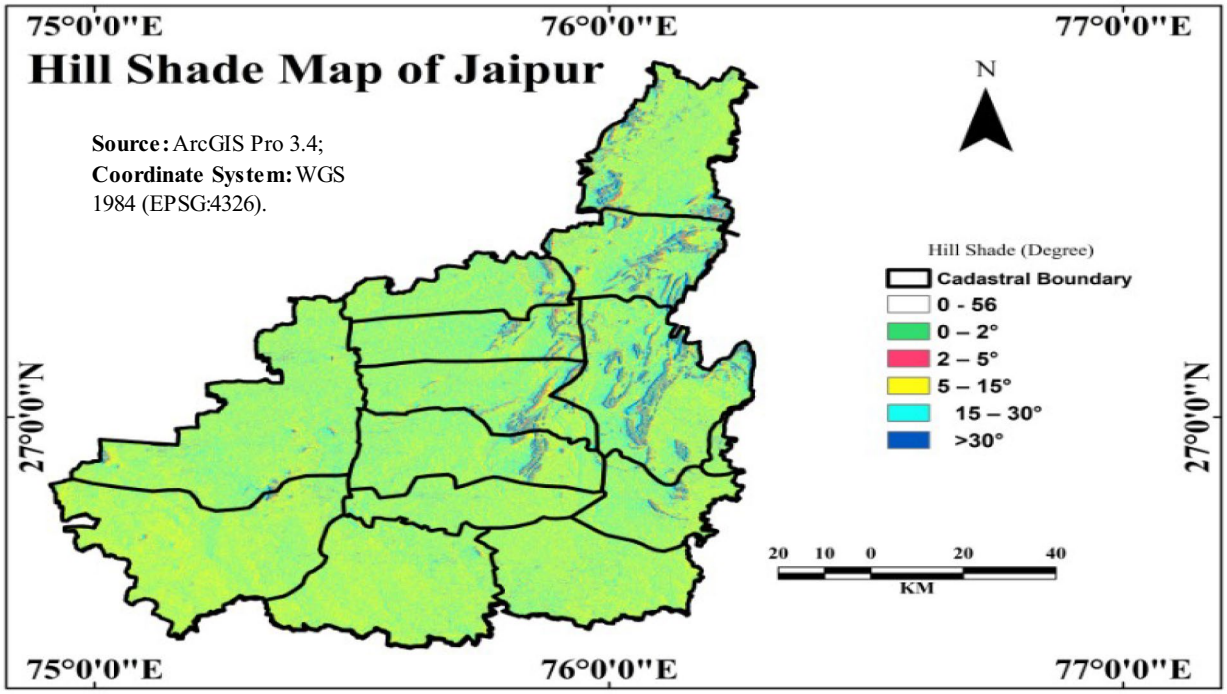


Fig. 11 Hill shade map used to identify suitable RWH zones in Jaipur District, Rajasthan.

Lineament density

Lineament density represents the spatial concentration of geological discontinuities, such as faults, fractures, and joints, which often enhance secondary porosity and permeability, thereby facilitating groundwater movement, infiltration, and subsurface storage. In Jaipur district, particularly along the Aravalli hill ranges, areas characterized by moderate to high lineament density are considered favorable locations for groundwater recharge and the development of rainwater harvesting structures. The lineament network was delineated from satellite imagery using remote sensing techniques and edge-enhancement filters[60], and the resulting lineament density map was incorporated as a hydrogeological predictor in the suitability assessment. Previous studies have demonstrated that regions with higher lineament density generally exhibit greater groundwater recharge potential and improved hydrological connectivity, making this variable an important indicator in rainwater harvesting and groundwater prospect mapping [61,62].

Land use/land cover (LULC)

LULC classes in Jaipur were derived from Landsat 8 satellite imagery, categorized into urban built-up, agricultural land, vegetation, open scrubland, barren areas, and water bodies. Impervious surfaces such as rooftops in urban zones offer higher runoff coefficients, making them efficient for rooftop RWH systems[63]. Agricultural and vegetative lands require soil-percolating systems such as percolation tanks and trenches. LULC thus plays a critical role in designing site-specific RWH strategies (NRAA, 2016) as shown in Fig. 13.Fig. 3 illustrating elevation map used to identify suitable RWH zones in Jaipur District, Rajasthan. The process includes data collection, reclassification, AHP-based weighted overlay, machine learning model training and validation, and generation of the final RWH suitability map.

Figure 4 illustrating TWI map used to identify suitable RWH zones in Jaipur District, Rajasthan. The process includes data collection, reclassification, AHP-based weighted overlay, machine learning model training and

validation, and generation of the final RWH suitability map. Figure 5 illustrating Slope map to identify suitable RWH zones in Jaipur District, Rajasthan. The process includes data collection, reclassification, AHP-based weighted overlay, machine learning model training and validation, and generation of the final RWH suitability map.

Figure 6 Map illustrating Aspect map used to identify suitable RWH zones in Jaipur District, Rajasthan. The process includes data collection, reclassification, AHP-based weighted overlay, machine learning model training and validation, and generation of the final RWH suitability map. Figure 7 illustrating Drainage Density map used to identify suitable RWH zones in Jaipur District, Rajasthan. The process includes data collection, reclassification, AHP-based weighted overlay, machine learning model training and validation, and generation of the final RWH suitability map.

Figure 8 illustrating Distance from Road map to identify RWH zones in Jaipur District, Rajasthan. The process includes data collection, reclassification, AHP-based weighted overlay, machine learning model training and validation, and generation of the final RWH suitability map. Figure 9 illustrating Rainfall map used to identify suitable RWH zones in Jaipur District, Rajasthan. The process includes data collection, reclassification, AHP-based weighted overlay, machine learning model training and validation, and generation of the final RWH suitability map.

Figure 10 illustrating Soil Profiles used to identify suitable RWH zones in Jaipur District, Rajasthan. The process includes data collection, reclassification, AHP-based weighted overlay, machine learning model training and validation, and generation of the final RWH suitability map. Figure 11 illustrating Hill Shade map to identify suitable RWH zones in Jaipur District, Rajasthan. The process includes data collection, reclassification, AHP-based weighted overlay, machine learning model training and validation, and generation of the final RWH suitability map.

Figure 12 illustrating Lineament Density map to identify suitable RWH zones in Jaipur District, Rajasthan. The process includes data collection, reclassification, AHP-based weighted overlay, machine learning model training and validation, and generation of the final RWH suitability map. Figure 13 illustrating LULC map to identify

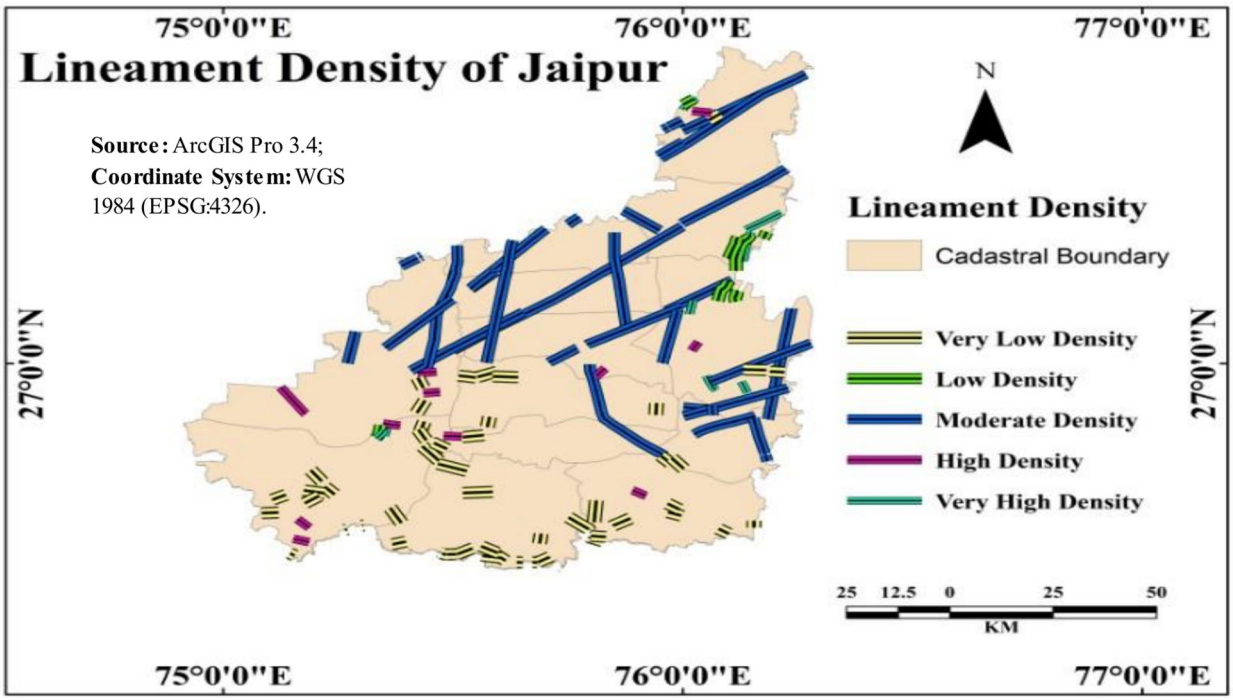


Fig. 12 Lineament density map to identify suitable RWH zones in Jaipur District, Rajasthan.

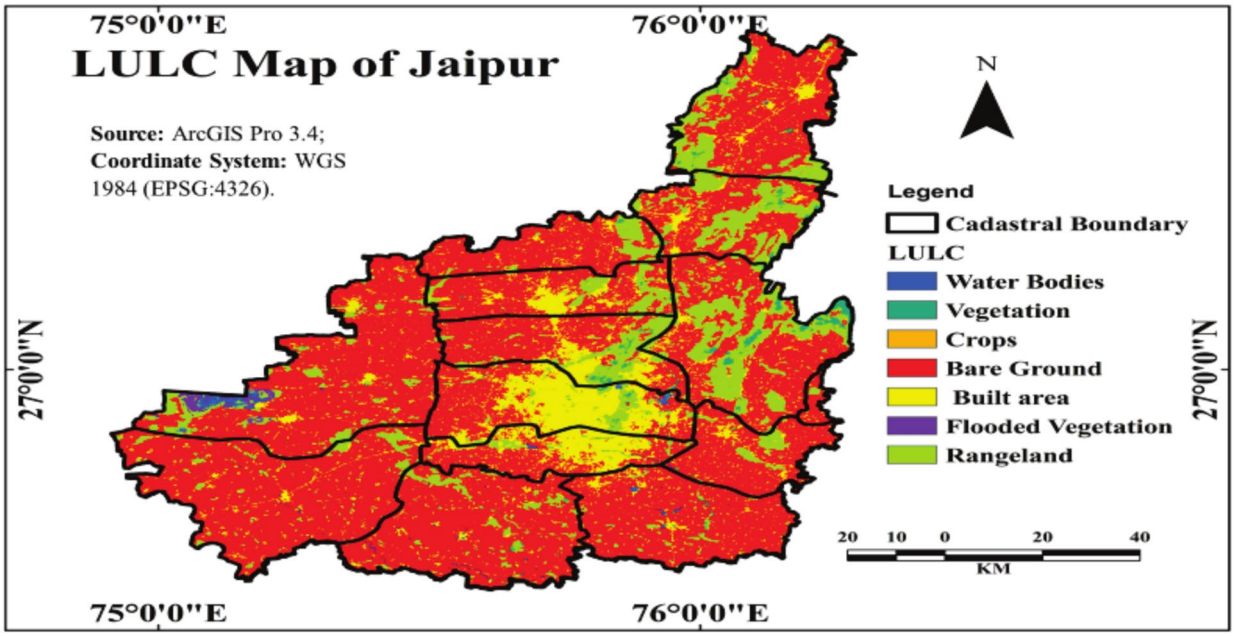


Fig. 13 LULC map used to identify suitable RWH zones in Jaipur District, Rajasthan.

suitable RWH zones in Jaipur District, Rajasthan. The process includes data collection, reclassification, AHP-based weighted overlay, machine learning model training and validation, and generation of the final RWH suitability map.

Analytical hierarchy process (AHP) for weight assignment

To determine the relative importance of each thematic parameter, the Analytical Hierarchy Process (AHP), as introduced by Saaty (1977), was employed. These multi-criteria decision-making method relies on pairwise comparisons to establish a hierarchy of influence among factors impacting RWH suitability.

Each parameter was compared using Saaty's 9-point fundamental scale:

1. 1 = Equal importance
2. 3 = Moderate importance
3. 5 = Strong importance
4. 7 = Very strong importance
5. 9 = Extreme importance
6. Values of 2, 4, 6, and 8 reflect intermediate judgments.
7. Reciprocal values (1/2 to 1/9) indicate inverse importance.

A pairwise comparison matrix was created, from which a priority vector (weights) was computed using the eigenvector method. The Consistency Index (CI) was calculated to ensure logical coherence in comparisons as given in Eq. 2:

$$CI = \frac{\lambda_{max} - n}{n - 1} \tag{2}$$

where:

- λ_{max} = Maximum eigenvalue of the pairwise comparison matrix.
- n = Number of criteria (order of the matrix).

To evaluate consistency, the Consistency Ratio (CR) was derived as given in Eq. 3:

$$CR = \frac{CI}{RI} \tag{3}$$

where:

• CI = Consistency Index.

Where RI is the Random Index, varying with matrix order (Saaty, 1980). A CR < 0.10 indicates acceptable consistency. The pairwise comparison matrix, normalized criterion weights, maximum eigenvalue λmax, consistency index (CI), random index (RI), and consistency ratio (CR) employed in the AHP framework are presented in Supplementary Tables S1 and S2. The computed consistency ratio (CR = 0.060) was substantially lower than the recommended threshold of 0.10, indicating acceptable judgment consistency and confirming the robustness and internal reliability of the weighting scheme[64]. Following consistency verification, the derived criterion weights were normalized and integrated with the corresponding thematic raster layers at a spatial resolution of 30 m × 30 m. Each thematic layer was subsequently reclassified according to its relative contribution to rainwater harvesting (RWH) potential. The weighted overlay analysis (WOA) technique was then implemented within a GIS environment to synthesize the reclassified layers and generate a composite RWH suitability map for Jaipur district[65]. To enhance the reliability and practical applicability of the resulting suitability framework, the model outputs were further evaluated through field-based validation and expert consultation, ensuring their suitability for both urban and rural hydrological settings within the study area.

Machine learning (ML) algorithms

This study employed four advanced supervised machine learning (ML) algorithms Extreme Gradient Boosting (XGBoost), Support Vector Machines (SVM), Gradient Boosted Machines (GBM), and CatBoost to model the spatial probability of rainwater harvesting (RWH) zones in the semi-arid terrain of Jaipur, Rajasthan. These algorithms were selected based on their proven performance in classification tasks involving complex non-linear relationships, heterogeneous feature sets, and spatially diverse data structures. Each algorithm was trained and tested using multi-criteria thematic layers to derive RWH suitability maps.Model performance was assessed using an independent testing dataset through multiple validation measures, including ROC-AUC, confusion matrix-derived overall accuracy, precision, recall, F1-score, and Cohen's Kappa coefficient.

Extreme gradient boosting (XGBoost)

XGBoost is an optimized gradient boosting framework designed to enhance model generalization and scalability[66]. Built on decision tree ensembles, XGBoost employs a gradient boosting approach where new trees are iteratively added to correct the residual errors of previous trees.

In XGBoost, the objective function is defined as given in Eq. 4:

$$\mathcal{L}(\phi) = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \tag{4}$$

$$\Omega(f_k) = \gamma T + \frac{1}{2}\lambda \parallel w \parallel^2 \tag{5}$$

where:

1.l denotes a differentiable convex loss function (e.g., logistic or squared error),

2.$\Omega(fk) = \gamma T + 12\lambda \parallel w \parallel 2\Omega(fk) = \gamma T + 21\lambda \parallel w \parallel istheregularizationterm$,

3.T is the number of leaves, and w the leaf weights.

XGBoost regularization mechanism significantly reduced overfitting, and its ability to handle missing values and high-dimensional datasets rendered it suitable for RWH site classification[67] .

Support vector machines (SVM)

Support Vector Machines (SVM) are supervised learning models that have been widely applied for binary and multi-class classification problems in hydrological and environmental domains[68,69] . An SVM forms one or several decision surfaces within a high-dimensional feature domain to divide categories while achieving greatest margin. The decision function as given in Eq. 5:

$$f(x) = \text{sign}\left(\sum_{i=1}^{n} \alpha_i y_i K(x_i, x) + b\right) \tag{6}$$

1. Where:
2. αi are Lagrange multipliers,
3. yi are class labels,
4. K(xi,x) is a kernel function (e.g., radial basis function),
5. *b* is the bias term.

The kernel trick enabled SVM to classify RWH suitability across non-linear feature spaces derived from land use, slope, and hydrology variables.

Gradient boosted machines (GBM)

Gradient Boosted Machines (GBM) function by training a series of weak learners typically decision trees in a sequential manner to minimize a specified loss function using gradient descent optimization[70,71]. The GBM model captures complex variable interactions and is capable of modelling non-linear relationships among environmental predictors.

The update rule at the m-th stage of boosting is as given in Eq. 6:

$$F_m(x) = F_{m-1}(x) + \nu \cdot h_m(x) \tag{7}$$

where:

1. Fm − 1(x) is the previous prediction,
2. $hm(x)$ is the new weak learner trained on the residuals,
3. v is the learning rate.

GBM's robustness to overfitting and capacity to deal with multicollinearity made it suitable for modelling RWH probabilities across spatially distributed thematic inputs.

CatBoost

CatBoost is a gradient boosting decision tree algorithm specifically optimized for handling categorical features, reducing prediction shift, and accelerating training performance[72–75] . It employs Ordered Boosting to eliminate target leakage and overfitting, a common problem in conventional boosting techniques.

The model builds upon the conventional boosting framework but encodes categorical features using an efficient permutation-driven encoding and handles missing data internally. Its objective function follows the general form as given in Eq. 7:

$$\mathcal{L} = \sum_{i=1}^{n} L\left(y_i, f\left(x_i\right)\right) + \Omega\left(f\right) \tag{8}$$

1. L: Overall objective (loss) function.
2. *yi*: True value for the i*i*-th instance.
3. *f*(*xi*): Predicted value from the model for the i*i*-th input.
4. Ω(*f*): Regularization term controlling model complexity.
5. *n*: Number of training instances.

CatBoost's high precision and minimal need for feature engineering made it particularly advantageous for multi-layered spatial RWH datasets with categorical inputs such as land cover class, soil Profile, or proximity to settlements.

Model validation and accuracy assessment

Evaluation of how effectively machine learning algorithms delineated RWH suitability regions was conducted through analysis of the Area Beneath the Receiver-Operating Characteristic curve, denoted AUC-ROC. The ROC curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR), with the AUC representing a scalar measure of overall model accuracy.

Interpretation of AUC values:

1. AUC ≤ 0.5: Performance equivalent to random classification
2. 0.7 ≤ AUC < 0.9: Acceptable to excellent predictive power
3. AUC ≥ 0.9: Highly accurate and reliable prediction

The inventory dataset was randomly partitioned into training (70%; n = 105) and testing (30%; n = 45) subsets. The training dataset was used for model calibration, hyperparameter optimization, and cross-validation, whereas the independent testing dataset was reserved exclusively for final model evaluation. To reduce over fitting and improve model generalization, ten-fold cross-validation was applied exclusively to the training dataset..

Results

Correlation matrix and multicollinearity assessment

Both elevation and slope were retained as predictor variables because they represent distinct but complementary geomorphological controls on rainwater-harvesting (RWH) suitability. Elevation influences regional topographic gradients, drainage organization, and water movement, whereas slope directly affects runoff velocity, infiltration opportunity, and the feasibility of water-retention structures[76]. Elevation and slope exhibited a moderate positive correlation (r = 0.584), reflecting the underlying topographic organization of the study area. Drainage density showed a moderate negative correlation with slope (r = −0.441), indicating systematic variation in drainage-network development across different terrain conditions. These relationships remained sufficiently moderate to justify retaining the variables as independent predictors in the RWH suitability assessment (Table 2).Land use/land cover (LULC) showed a weak positive correlation with soil profile (r = 0.321) and a moderate positive correlation with curve number (CN; r = 0.479), indicating spatial associations among land-surface characteristics, soil conditions, and runoff response. Soil profile exhibited a weak negative correlation with slope (r = −0.296), suggesting that soil characteristics vary systematically along the topographic gradient. The Topographic Wetness Index (TWI) was moderately negatively correlated with slope (r = −0.501), indicating a greater tendency for

Table 2 Evaluation of multicollinearity among environmental variables influencing rainwater harvesting.

Parameters	Tolerance	VIF
Aspect	0.782	1.279
Drainage Density	0.837	1.195
Elevation	0.427	2.343
Geology	0.726	1.378
Hill shade	0.752	1.329
Lineament Density	0.775	1.291
LULC	0.786	1.272
Rainfall	0.866	1.154
Distance from Road	0.597	1.675
Slope	0.376	2.659
TWI	0.676	1.480

water accumulation and moisture retention in relatively gentle terrain. CN exhibited a strong positive correlation with runoff potential (r = 0.737), which is hydrologically consistent because higher CN values generally represent greater runoff generation and lower infiltration or abstraction potential.

Multicollinearity among the predictor variables was further assessed using the variance inflation factor (VIF) and tolerance statistics. All predictors had VIF values below the conservative threshold of 5.0 and tolerance values above 0.20, indicating the absence of problematic multicollinearity. Slope and CN exhibited the highest VIF values of 2.891 and 2.621, with corresponding tolerance values of 0.346 and 0.381, respectively. Drainage density, soil profile, and LULC showed comparatively lower VIF values ranging from 1.143 to 1.894, with tolerance values between 0.528 and 0.874. Overall, the correlation and multicollinearity diagnostics demonstrate that the selected thematic variables provide sufficiently distinct information for RWH suitability modeling, with no substantial redundancy likely to compromise model stability or interpretation (Fig. 14).

Figure 14 illustrating correlation matrix illustrates the strength and direction of relationships among the eleven parameters considered for RWH suitability analysis, namely, Aspect, Drainage Density, Elevation, Geology, Hillshade, Lineament Density, LULC, Rainfall, Distance from Road, Slope, and (TWI). Correlation coefficients range from + 1 (indicating a perfect positive relationship) to -1 (indicating a perfect negative relationship), as depicted on the scale adjacent to the main figure. The strongest positive correlation was observed between Slope and Elevation, suggesting that steeper gradients correspond to higher altitudes. Conversely, the strongest negative correlation occurred between Slope and TWI, indicating that areas with steeper slopes tend to exhibit lower moisture accumulation potential.

Rainwater harvesting suitability zonation

Five predictive approaches, namely Extreme Gradient Boosting (XGBoost), CatBoost, Support Vector Machine (SVM), Gradient Boosting Machine (GBM), and Analytical Hierarchy Process (AHP), were applied and compared to delineate rainwater-harvesting (RWH) suitability zones across the semi-arid landscape of Jaipur, Rajasthan. Each model captured spatial variations in RWH suitability associated with differences in topography, hydrological characteristics, soil conditions, and land-cover patterns. For consistent spatial interpretation, the continuous suitability outputs were classified using the Natural Breaks (Jenks) method in ArcGIS 10.8 into five categories: very low, low, moderate, high, and very high suitability.

The GBM model classified 1,011.3 km2 (9.9%) of the study area as very low suitability and 1,367.7 km2 (13.4%) as low suitability. Moderate suitability represented the largest class, covering 3,498.0 km2 (34.2%), followed by high suitability at 2,457.4 km2 (24.0%) and very high suitability at 1,885.6 km2 (18.5%). The spatial pattern indicated a gradual transition from relatively unsuitable areas to more favorable RWH zones in locations characterized by improved runoff-generation, storage, and infiltration conditions[77].

CatBoost produced a comparatively concentrated distribution of moderate suitability. Very low and low suitability accounted for 875.6 km2 (8.6%) and 1,186.4 km2 (11.6%), respectively, whereas moderate suitability

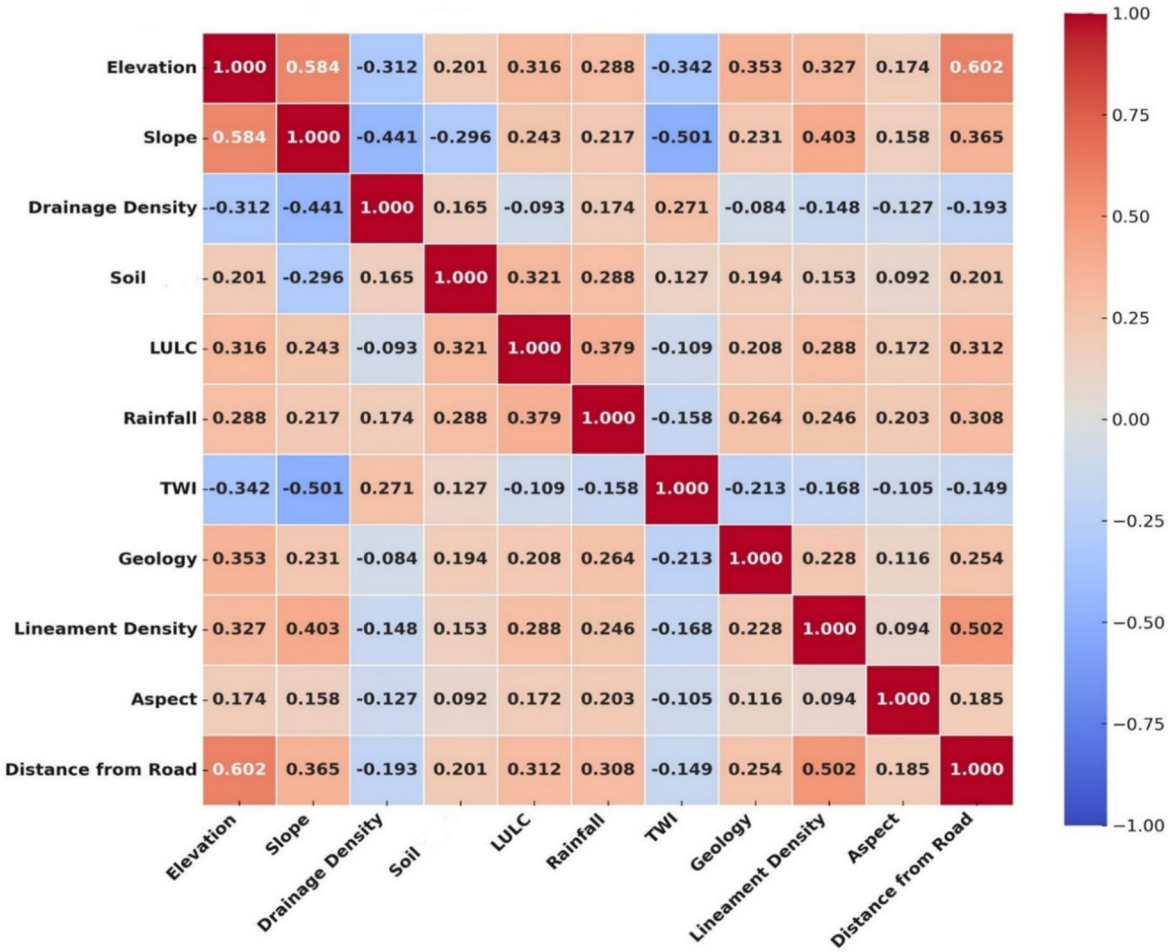


Fig. 14 The correlation matrix illustrates the strength and direction of relationships among the eleven parameters considered for RWH suitability analysis.

occupied 3,985.0 km2 (39.0%). High suitability covered 2,484.0 km2 (24.3%), while 1,689.0 km2 (16.5%) was classified as very highly suitable. Collectively, the high and very high classes represented 40.8% of the modelled area, demonstrating the substantial extent of potentially favourable RWH locations identified by CatBoost[78].

The SVM model generated a relatively heterogeneous spatial distribution of suitability classes. Very low suitability covered 1,282.9 km2 (12.6%), while low suitability accounted for 1,421.6 km2 (13.9%). Moderate suitability remained dominant, occupying 3,614.2 km2 (35.4%). High and very high suitability covered 2,179.3 km2 (21.3%) and 1,722.0 km2 (16.8%), respectively[79]. These results indicate that SVM identified favourable RWH locations across broader topographic and hydrological transition zones (Fig. 15).

XGBoost classified approximately 1,140.5 km2 (11.2%) as very low suitability and 1,643.2 km2 (16.1%) as low suitability. Moderate suitability represented the largest class at 3,533.7 km2 (34.6%), followed by high suitability at 2,057.8 km2 (20.1%). Approximately 18.0% of the study area was classified as very highly suitable. Spatially, the high and very high suitability zones were concentrated predominantly toward the southern part of the study area, reflecting the combined influence of favourable terrain, drainage, soil, land-cover, and rainfall conditions[13].

The AHP model exhibited a more generalized spatial pattern than the machine-learning models. Moderate suitability was the dominant category, covering 5,150.0 km2 (50.4%) of the effective modelled area. High suitability accounted for 2,211.0 km2 (21.6%), while very high suitability covered 1,561.0 km2 (15.3%). In contrast, low

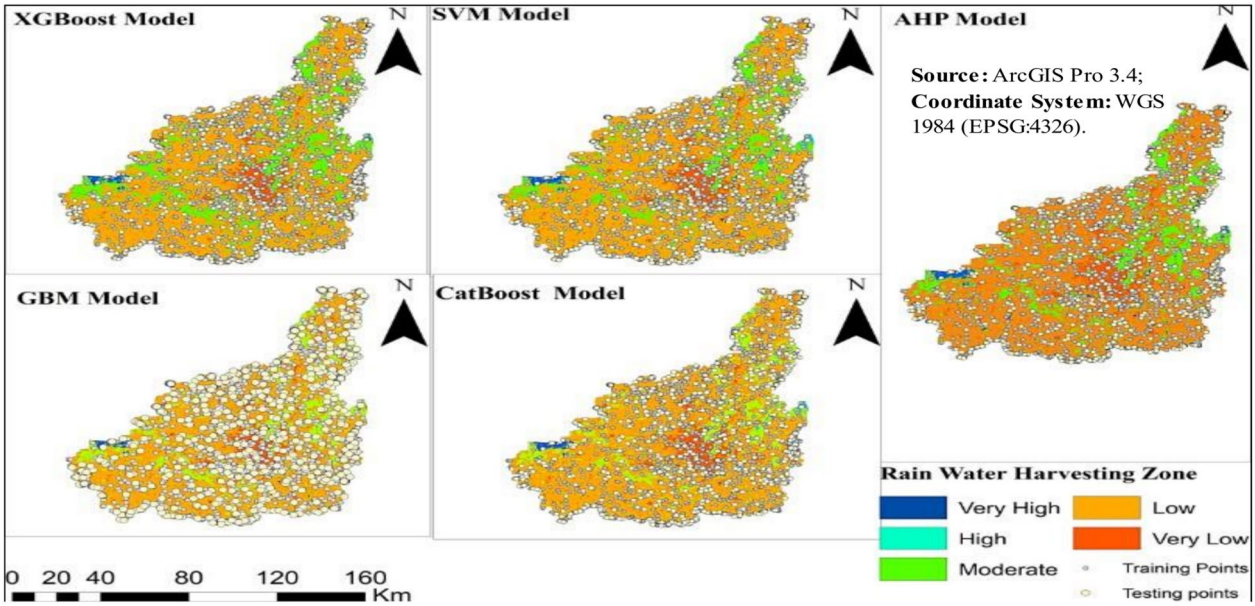


Fig. 15 Rainwater harvester zone maps derived from varied algorithms (XGBOOST, SVM, GBM, CATBOOST, AHP) models.

and very low suitability occupied only 978.0 km2 (9.6%) and 320.0 km2 (3.1%), respectively. The larger proportion of moderate suitability generated by AHP indicates a comparatively smoother differentiation of RWH potential, whereas the machine-learning models produced greater spatial discrimination among suitability classes[28].

Following spatial masking and preprocessing, the effective area considered for modelling was approximately 10,220 km2, compared with the broader geographical extent of approximately 11,000 km2. Across all five models, moderate suitability consistently represented a substantial proportion of the study area. However, clear differences were observed in the extent of the extreme suitability classes. GBM identified the largest combined high and very high suitability area (42.5%), followed by CatBoost (40.8%), SVM (38.1%), XGBoost (38.1%), and AHP (36.9%). Conversely, AHP produced the smallest combined low and very low suitability area (12.7%), indicating less pronounced separation of unsuitable locations compared with the machine-learning approaches.

Despite differences in class proportions, the five models exhibited broad spatial agreement in identifying the principal RWH suitability pattern across Jaipur. Comparatively lower suitability was concentrated mainly in the northern sector, whereas the central region was predominantly characterized by moderate suitability. High to very high suitability zones were more consistently identified toward the southern sector, where favorable combinations of hydrological, geomorphological, and land-surface conditions enhanced RWH potential (Fig. 16). Differences among model outputs are expected because each approach represents predictor relationships through different mathematical structures and optimization procedures. Nevertheless, the recurring identification of priority zones across independent modelling approaches strengthens confidence in the spatial pattern of RWH suitability and provides a basis for targeting field verification and future RWH interventions in the most promising areas.

Model validation and predictive performance assessment

The forecasting performance of five classification algorithms: XGBoost, SVM, GBM, CatBoost, and AHP has been comprehensively evaluated using the main evaluation indicator Area Under the Receiver Operating Characteristic Curve (AUC). The AUC is a strong indicator of the discriminatory power of a model because it can capture its capacity to discriminate between appropriate and inappropriate zones for RWH at various thresholds[28,80].

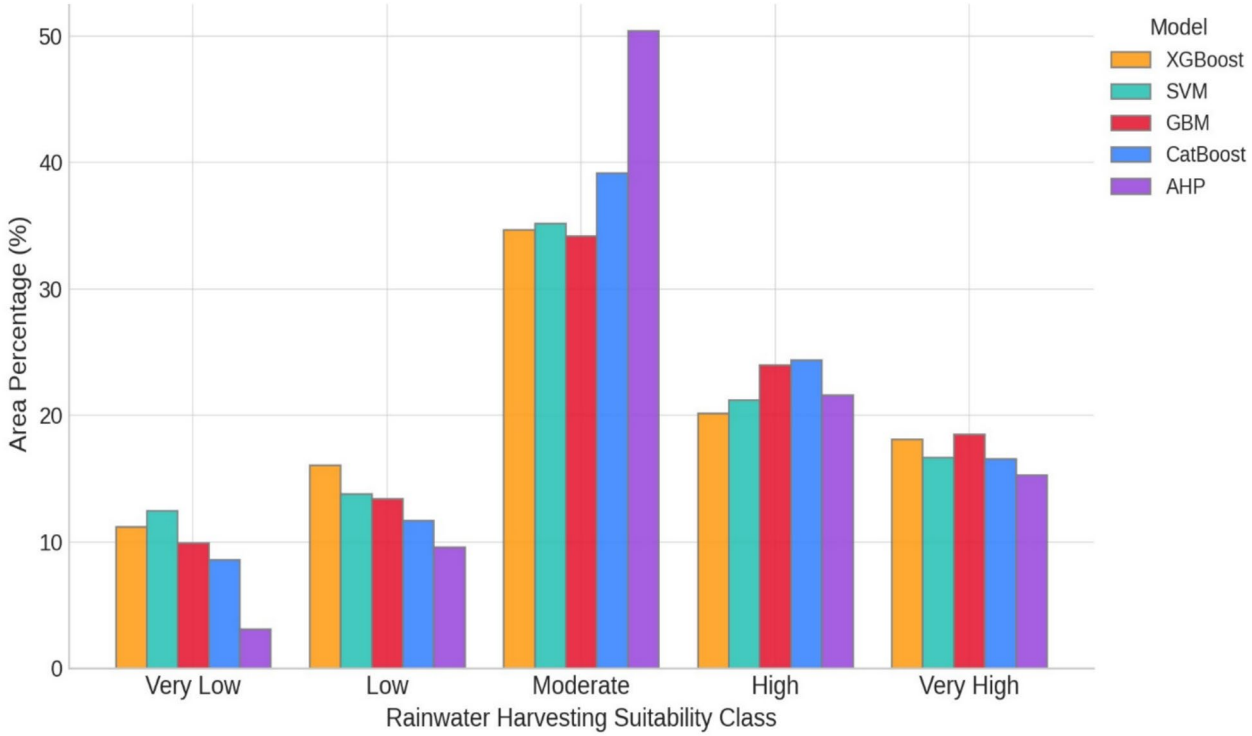


Fig. 16 An illustrative chart presents the proportion of land attributed to very high, low, moderate, high, and very high RWH suitability classes.

Table 3 Validation by area under the curve (AUC) of five Rainwater harvesting models.

Model	AUC
XGBoost	0.78
SVM	0.76
AHP	0.73
GBM	0.68
CatBoost	0.69

The XGBoost model presented the highest AUC value of 0.78 as shown in Table 3 and Fig. 17, suggesting its relatively strong classification capability and robustness to model the spatial distribution of RWH suitability.

SVM followed closely with an AUC of 0.76, demonstrating reliable performance in capturing spatial variability[81]. The CatBoost and GBM models recorded AUC values of 0.69 and 0.68, respectively, indicating moderate predictive strength. Despite being ensemble-based approaches, their performance was marginally lower, possibly due to overfitting or suboptimal hyperparameter configurations in the current dataset[82].The AHP model yielded an AUC value of 0.73, indicating moderate predictive performance. However, it was not the lowest-performing model, as GBM and CatBoost recorded lower AUC values of 0.68 and 0.69, respectively.The reported ROC-AUC and classification metrics were obtained using the independent testing dataset following model optimization through ten-fold cross-validation on the training data[82,83].These results clearly underscore the superior performance of data-driven machine learning techniques, particularly XGBoost and SVM, in handling high-dimensional, non-linear, and multi-criteria datasets. Their enhanced capability to learn from complex spatial patterns makes them more suitable for environmental modelling applications such as RWH site suitability analysis[84,85]. The findings advocate for a continued shift toward machine learning-based ensemble models over traditional methods like AHP in spatial prioritization frameworks (Table 3; Fig. 17).

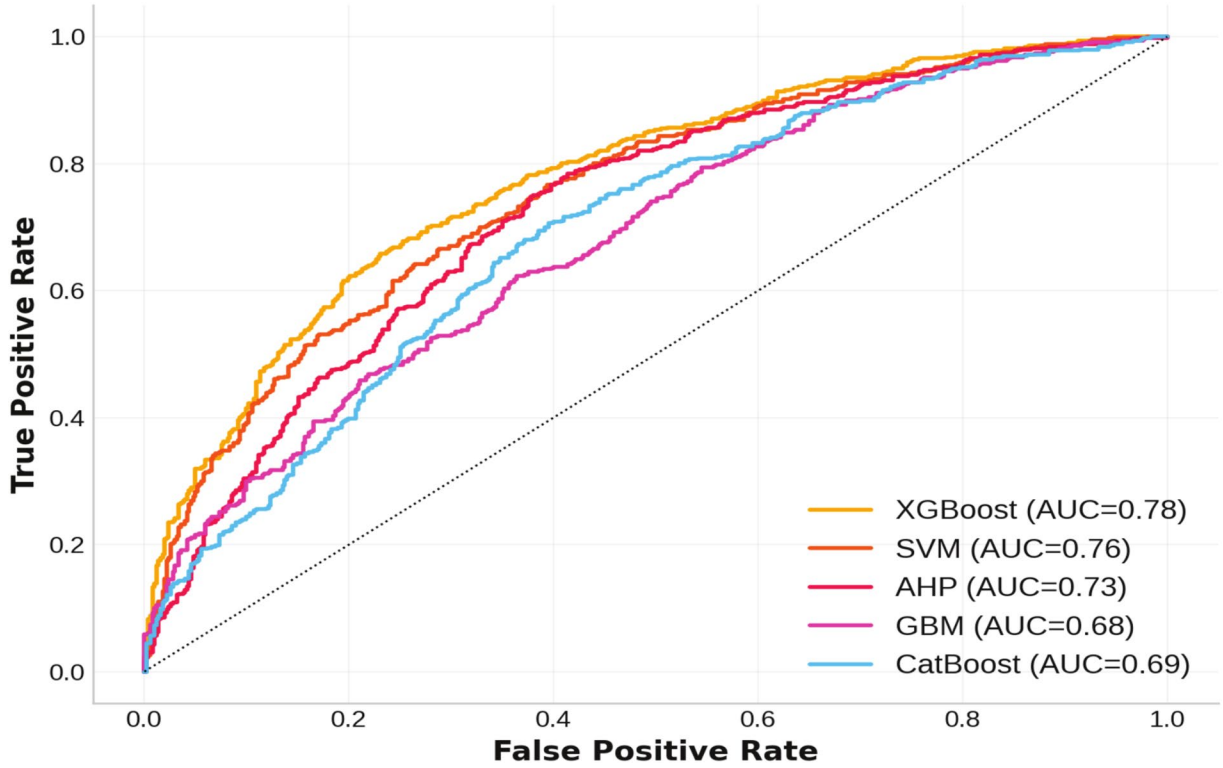


Fig. 17 The area under the curve metric.

Importance of conditioning variables in RWH prediction

The spatial delineation of RWH suitability has been governed by an ensemble of environmental and topographic conditioning variables. In the present investigation, eleven key parameters were integrated across four advanced machine learning algorithms XGBoost, CatBoost, SVM, and GBM alongside a geospatial decision-support technique, namely the AHP. The relative significance of each variable was quantitatively assessed based on the outputs of the two best-performing machine learning models, namely XGBoost and SVM, which achieved the highest predictive accuracies with AUC values of 0.78 and 0.76, respectively[13].Variable importance analysis derived from the XGBoost and SVM models revealed that rainfall, drainage density, elevation, and topographic wetness index were the most influential factors controlling the spatial distribution of rainwater harvesting suitability across Jaipur district (Fig. 18).The analysis revealed marked variability in variable contribution patterns between GBM and XGBoost in the context of the physio graphically complex terrain of Jaipur, Rajasthan. The GBM algorithm displayed a strong dependence on a limited set of predictors, particularly drainage density and elevation, which contributed 22.810% and 22.040%, respectively, to the overall model efficacy. These outcomes suggest that GBM predominantly relied on hydrological and altitudinal gradients, reflecting the dominance of surface runoff dynamics and elevation-induced flow accumulation in influencing RWH potential. While such dependence can enhance localized accuracy, it may simultaneously compromise model generalizability, particularly in the presence of variable data inconsistencies or geospatial resolution disparities[86].

Conversely, the XGBoost model exhibited a broader and more balanced distribution of importance across multiple features, enhancing its adaptability to varied ecological and geomorphic contexts[87]. Rainfall emerged as the principal predictor, accounting for 15.730% of the model's explanatory power, followed by drainage density (15.550%), elevation (11.450%), aspect (9.350%), TWI (8.820%), and slope (8.340%). This diversified contribution structure indicates XGBoost's capability to capture complex, non-linear interdependencies among terrain attributes, hydroclimatic gradients, and spatial heterogeneity[88,89]. The juxtaposition of these findings underscores

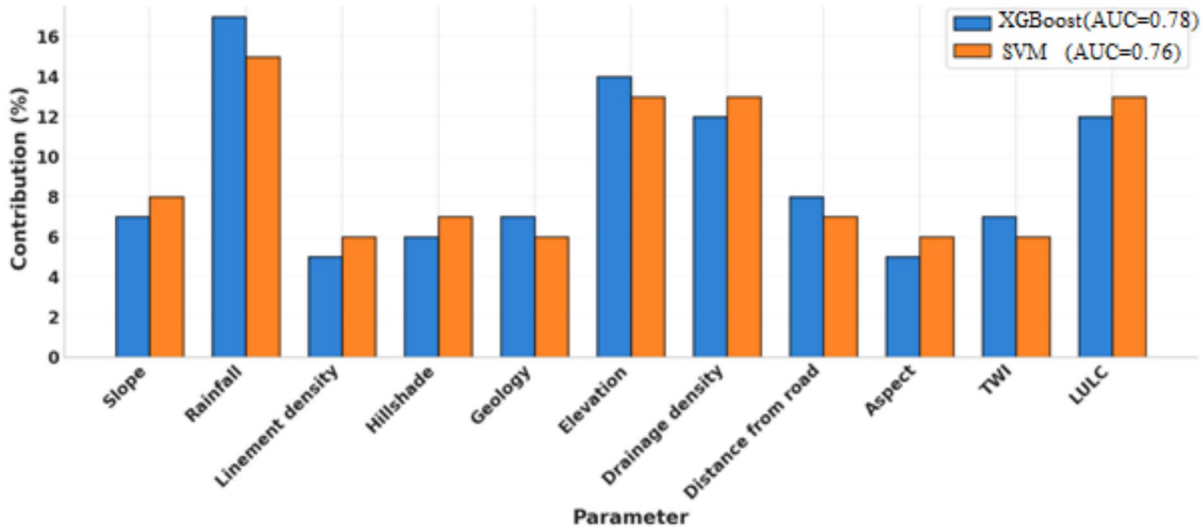


Fig. 18 Relative contribution of individual parameters to the identification of RWH suitability zones using the XGBoost and SVM models.

the methodological divergence between the two algorithms. GBM, while efficient in variable-specific contexts, may demonstrate lower resilience under data variability. In contrast, XGBoost integrates a more resilient and generalizable modelling architecture, rendering it more suitable for heterogeneous terrain analysis and decision-making in semi-arid hydrological planning [90,91]. This multi-variable engagement enhances predictive robustness and supports the model's scalability for broader environmental applications, including climate-resilient water resource management and adaptive watershed interventions. The maps shown in Fig. 15 depicts rainfall-capture suitability areas derived from varied algorithms (XGBOOST, SVM, GBM, CATBOOST, AHP) models. These regions were grouped as very low, low, moderate, high, and very high capability categories levels. The yellow and black markers across every diagram denote the training and validation samples used for algorithmic forecasting tasks.

The five algorithms (XGBOOST, SVM, GBM, CATBOOST, and AHP) applied to delineate RWH classes estimated the share of terrain falling within each category. Each method partitioned the region into five RWH suitability classes based on eleven criteria. Distinct hues represent the various suitability classes across the region (Fig. 16).

The spatial patterns of the suitability classes of rainwater harvesting (RWH) as determined by various models indicate both similar and significant variations within the study area (Table 4).

Moderate suitability class was present in a significant portion of the study area for all evaluated models. The AHP model assigned the highest percentage of the study area to the moderate suitability category (50.4%), which was significantly higher than the machine-learning model results, which ranged from 34.2% to 39.0%. The largest moderate suitability zone (39.0%) was obtained with CatBoost among machine-learning methods, followed by SVM (35.4%), XGBoost (34.6%), and GBM (34.2%). The relatively high proportion of moderate class produced by AHP can be explained by the weighting structure of the expert-driven classification, where fixed weights of criteria and class scores are applied throughout the study area. This method can reduce local spatial differences and create larger intermediate zones. Machine-learning algorithms can, in contrast, detect the nonlinear relationship and interactions between environmental predictors and produce more distinctiveness among suitability classes. Hence, the observed differences should be seen as effects of the different mathematical structures and decision mechanisms of the models and not merely as overestimation by AHP[92,93].

Notable inter-model differences were also observed in the high and very high suitability categories, which are particularly important for prioritizing practical RWH interventions. The machine-learning models classified

Table 4 Area (km^2) and corresponding percentage (%) of each RWH suitability class as determined by the XGBOOST, SVM, GBM, CATBOOST, and AHP.

Class	XGBoost Area	XGBoost %	SVM Area	SVM %	GBM Area	GBM %	CatBoost Area	CatBoost %	AHP Area	AHP %
Very Low	1140.5	11.2	1282.9	12.6	1011.3	9.9	875.6	8.6		
Low	1643.2	16.1	1421.6	13.9	1367.7	13.4	1186.4	11.6		
Moderate	3533.7	34.6	3614.2	35.4	3498	34.2	3985	39		
High	2057.8	20.1	2179.3	21.3	2457.4	24	2484	24.3		
Very High	1844.8	18	1722	16.8	1885.6	18.5	1689	16.5		
Total	10,220	100	10,220	100	10,220	100	10,220	100	10,220	100
AHP values										
Very Low									320	3.1
Low									978	9.6
Moderate									5150	50.4
High									2211	21.6
Very High									1561	15.3

approximately 20.1–24.3% of the study area as highly suitable, with CatBoost (24.3%) and GBM (24.0%) producing the largest estimates, followed by SVM (21.3%) and XGBoost (20.1%). For the very high suitability category, GBM delineated the largest proportion (18.5%), followed by XGBoost (18.0%), SVM (16.8%), and CatBoost (16.5%). In comparison, the AHP model classified 21.6% and 15.3% of the study area as high and very high suitability, respectively. The relatively greater differentiation produced by the machine-learning models may reflect their ability to learn complex nonlinear relationships among rainfall, drainage density, elevation, TWI, slope, and other conditioning variables[94]. However, a larger predicted high-suitability area should not automatically be interpreted as superior model performance; rather, the reliability of these spatial predictions should be evaluated using independent validation metrics and statistical comparisons[30].

Differences were similarly evident in the low and very low suitability categories. SVM classified 12.6% and 13.9% of the study area as very low and low suitability, respectively, while XGBoost classified 11.2% and 16.1% in these categories. In contrast, AHP identified only 3.1% as very low and 9.6% as low suitability. This wider distribution across suitability classes suggests that data-driven models provide greater spatial discrimination of marginal, moderate, and highly favorable RWH zones, whereas AHP tends to concentrate a larger proportion of the landscape within the moderate category[95]. Nevertheless, AHP remains valuable as an interpretable knowledge-driven decision-support approach, particularly where extensive training data are unavailable[96]. Consequently, the complementary use of AHP and machine-learning models can provide a more comprehensive basis for RWH planning by combining expert knowledge with data-driven identification of complex spatial patterns[93].

Although XGBoost achieved the highest predictive performance, its improvement over the conventional AHP approach was moderate. Nevertheless, machine-learning algorithms offer several methodological advantages for rainwater harvesting (RWH) suitability assessment. In particular, they can capture complex and nonlinear relationships among environmental predictors, accommodate interactions among multiple conditioning factors, and reduce dependence on subjective expert-derived weighting schemes. These characteristics make machine-learning approaches flexible and data-driven alternatives for modelling spatial variations in RWH suitability[97]. The area under the receiver operating characteristic curve (ROC-AUC) was used to evaluate the discriminatory ability of the models to distinguish between suitable and unsuitable RWH locations. ROC-AUC values range from 0 to 1, with higher values indicating greater discriminatory performance and class separability. Among the five models evaluated, XGBoost achieved the highest AUC (0.78), followed closely by SVM (0.76). AHP produced an AUC of 0.73, whereas CatBoost (0.69) and GBM (0.68) showed comparatively lower discriminatory performance. These results indicate that XGBoost and SVM provided the strongest overall discrimination of RWH suitability across the semi-arid landscape of Jaipur, Rajasthan. As illustrated in Fig. 17, the ROC curves represent the trade-off between the true positive rate (sensitivity) and false positive rate (1-specificity) across classification thresholds, while the diagonal reference line represents discrimination equivalent to random classification. The

greater departure of the XGBoost and SVM curves from this reference line is consistent with their comparatively higher AUC values and stronger ability to differentiate favourable from unfavourable RWH locations[13].

To determine whether the observed differences in ROC-AUC represented statistically meaningful improvements rather than numerical differences alone, pairwise comparisons were conducted using DeLong's test[98]. No statistically significant difference was detected between XGBoost and SVM ($p=0.530$) or between XGBoost and AHP ($p=0.129$). In contrast, XGBoost performed significantly better than CatBoost ($p=0.015$) and GBM ($p=0.007$). Therefore, although XGBoost recorded the highest overall AUC (0.78), its numerical improvement over AHP (AUC = 0.73) cannot be interpreted as a statistically significant increase in discriminatory performance at the 0.05 significance level. This finding demonstrates that increased model complexity does not necessarily translate into a statistically significant improvement in predictive discrimination. Nevertheless, machine-learning approaches, particularly XGBoost and SVM, remain valuable because of their ability to capture nonlinear interactions and complex multivariate relationships among environmental factors while reducing reliance on subjective weighting procedures. XGBoost provided the strongest numerical predictive performance, but its advantage over AHP should be interpreted cautiously. Detailed pairwise ROC-AUC comparisons and associated significance statistics are presented in Supplementary Table S5.

The relative importance analysis was performed using the two best performing machine learning models which gave highest ROC-AUC values among the tested models as shown in Fig. 18, namely XGBoost and SVM. All eleven predictors—slope, rainfall, lineament density, hillshade, geology, elevation, drainage density, distance from roads, aspect, topographic wetness index (TWI), and land use/land cover (LULC) were included in the analysis to quantify the contribution of each predictor variable. The climatic and topographic variables were found to be the most significant in both the rainfall model and the elevation and drainage density models, suggesting the dominant role of these variables in the generation of surface runoff, water accumulation and the potential for groundwater recharge in the Jaipur district. The spatial variation in precipitation pattern, terrain elevation and drainage condition represent the highest influence on the suitability zone for RWHs in the semi-arid environment of Rajasthan as shown in Fig. 18. In addition to the ROC-AUC analysis, the model's predictive efficiency was validated using a confusion matrix-based metrics on the independent test data as described in. The overall classification accuracy of the highest model (Returned model) was 88.89%, with a Kappa coefficient of 0.64, indicating a satisfactory level of agreement between observed and predicted suitability classes for rain water harvesting. The confusion matrix also enabled the computation of various performance metrics including precision, recall, and f1-scores for each class, and hence contributed to a comprehensive assessment of the model reliability.

Spatial consensus analysis of RWH suitability

The spatial delineation of rainwater harvesting (RWH) potential across Jaipur district showed substantial agreement among the CatBoost, SVM, XGBoost, AHP, GBM, and Consensus models, as demonstrated by the comparative suitability maps and continuous wavelet power spectra (Fig. 19). Despite differences in model architecture and predictive mechanisms, all models exhibited broadly consistent spatial patterns. The southern and southwestern parts of Jaipur were predominantly classified within the high and very high suitability categories, whereas the northern and northeastern regions were mainly characterized by low to very low RWH suitability. This spatial variation can be attributed to the combined influence of topography, drainage characteristics, rainfall distribution, infiltration potential, and geological conditions. In particular, the southern plains are characterized by relatively gentle terrain, lower elevations, favorable runoff accumulation conditions, and suitable hydrogeological characteristics, which collectively enhance water retention and groundwater recharge potential[99].

The Consensus model, which integrates predictions from the individual modelling approaches, classified approximately 36.45% of the study area as high to very high suitability, 34.52% as moderately suitable, and 29.03% as low to very low suitability. The district-wide mean suitability index of 0.53, together with a coefficient of variation of 49.1%, indicates substantial spatial heterogeneity in RWH potential. Furthermore, Moran's I of 0.63 ($p<0.001$) demonstrates significant positive spatial autocorrelation, confirming that areas with similar

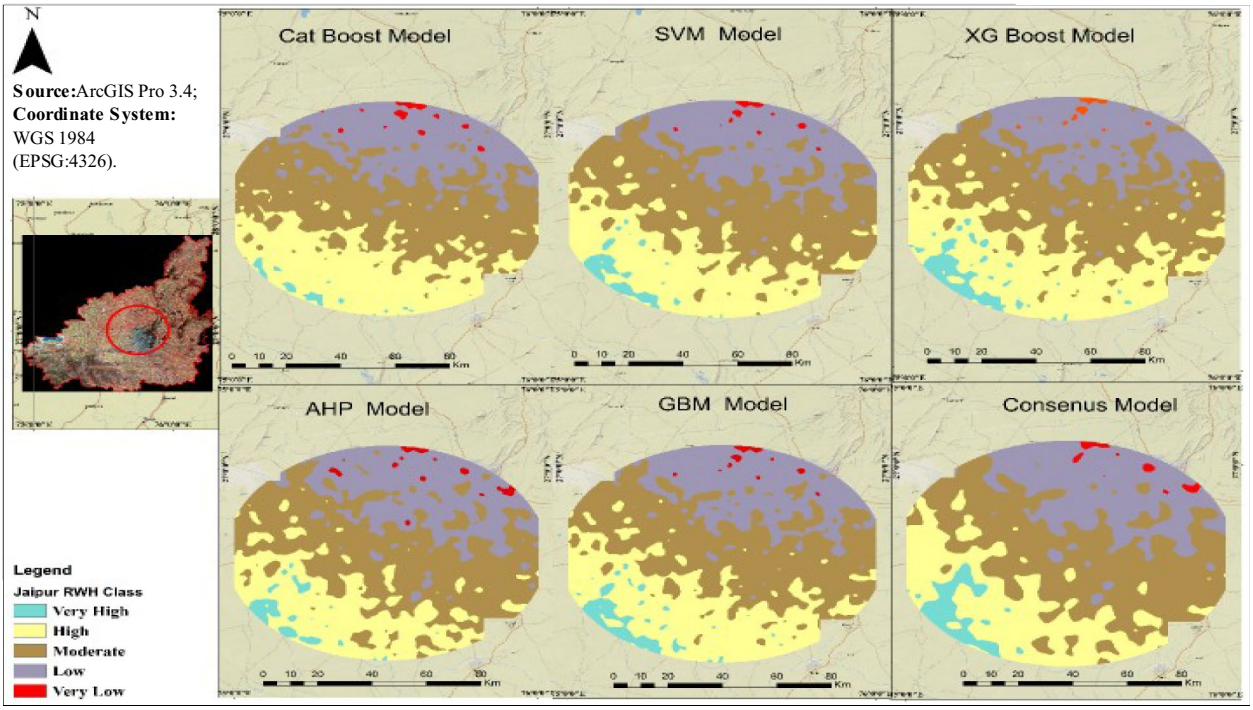


Fig. 19 Spatial consensus mapping of rainwater harvesting suitability in semi-arid Jaipur district, India. The inset map (red circle) shows the location of Jaipur district within Rajasthan, India.

suitability values are spatially clustered rather than randomly distributed[100]. Highly suitable zones were particularly concentrated within the southeastern lowlands and alluvial plains, reflecting the spatial organization of underlying physiographic and hydrological controls.

Continuous wavelet analysis further revealed the multi-scale spatial structure of the predicted RWH suitability patterns[93]. Across the models, dominant wavelet power was concentrated primarily at medium to large spatial scales of approximately 16–64 km, indicating that RWH suitability is influenced by broad-scale environmental and hydrological processes rather than exclusively by localized variations. Areas of high wavelet power, represented by red and orange regions in the spectra, were particularly evident at scales of approximately 32–64 km. These coherent high-power regions indicate spatially organized hydrological behavior associated with runoff generation, drainage configuration, infiltration processes, and groundwater recharge across Jaipur district (Fig. 20).

The wavelet signatures also revealed differences in the spatial structures captured by the individual models. XGBoost and SVM exhibited relatively strong and continuous high-power regions, indicating their greater ability to represent nonlinear relationships and multi-scale interactions among environmental predictors. The Consensus model displayed particularly prominent and spatially coherent power concentrations within the cone of influence (COI), suggesting that integration of multiple model predictions can reduce model-specific uncertainty and improve the robustness and spatial consistency of suitability estimates[101]. In contrast, the AHP model exhibited comparatively fragmented wavelet structures, potentially reflecting the limitations of fixed expert-derived weights in representing complex nonlinear interactions among environmental variables.

The global wavelet spectra further showed that dominant power across the models generally occurred at spatial scales greater than 32 km. This finding emphasizes the importance of regional-scale physiographic and hydrological controls, particularly rainfall gradients, topographic variability, drainage-network configuration, infiltration characteristics, and groundwater recharge conditions, in determining the spatial organization of RWH suitability[102]. The dominance of medium- to large-scale spatial structures also has important implications for RWH planning. Rather than relying exclusively on isolated site-level assessments, RWH interventions should initially

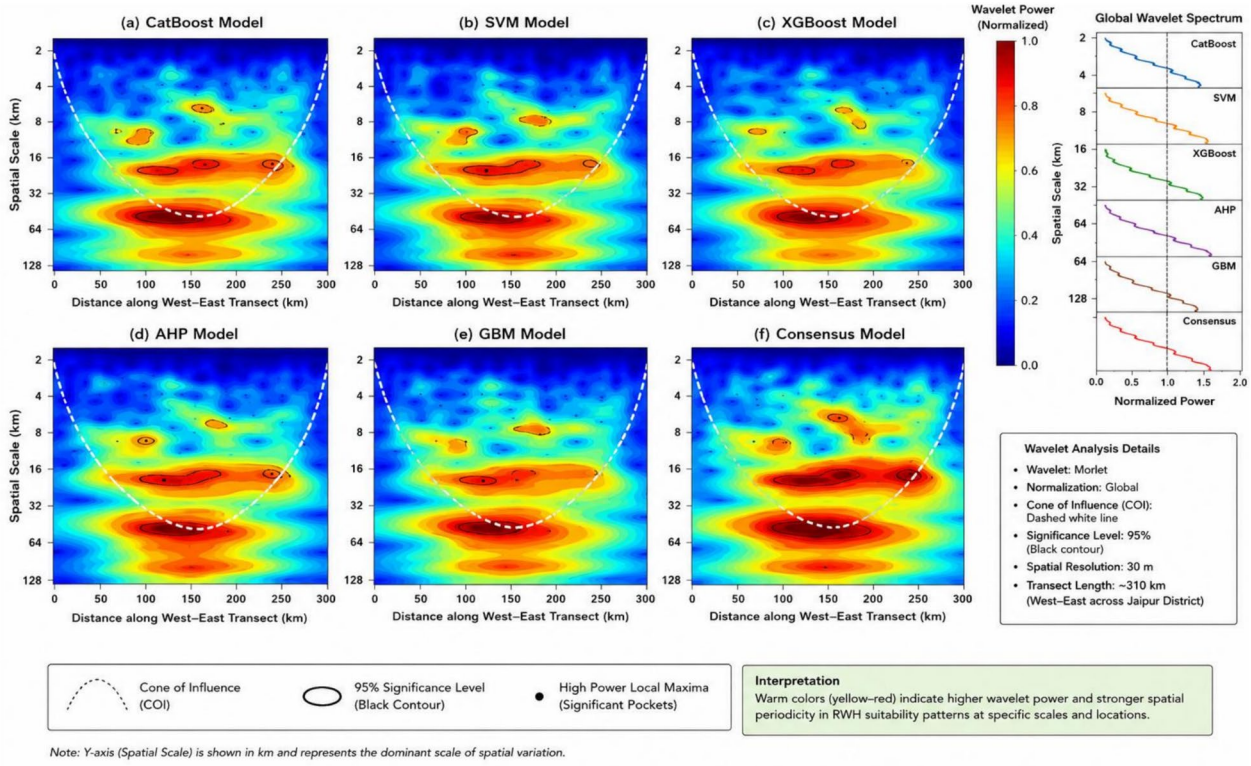


Fig. 20 Multi-scale wavelet analysis of rainwater harvesting suitability across machine learning and ensemble models in Jaipur District, India.

be prioritized at sub-basin and watershed scales, where broader hydrological connectivity and landscape controls can be incorporated into planning decisions. Detailed site-level investigations can subsequently be conducted to determine the precise locations and appropriate types of RWH structures within the identified priority zones.

Discussion

The present study demonstrates the feasibility of integrating machine-learning (ML) algorithms with geospatial techniques for delineating rainwater harvesting (RWH) suitability zones across Jaipur district, Rajasthan, a semi-arid region characterized by increasing water stress, rapid urbanization and substantial interannual rainfall variability. Comparative assessment of XGBoost, CatBoost, Support Vector Machine (SVM), Gradient Boosted Machine (GBM) and Analytical Hierarchy Process (AHP) provided a systematic basis for evaluating the relative strengths of data-driven and knowledge-driven approaches to spatial RWH assessment.

A distinct spatial gradient in RWH suitability was observed across the study area. The northern, central and southern sectors were predominantly characterized by low, moderate and high suitability, respectively. This spatial organization broadly reflects the geomorphological and hydroclimatic characteristics of Jaipur district. The northern sector is characterized by comparatively steep terrain, which can promote rapid runoff and reduce opportunities for infiltration and water retention. In contrast, the southern lowlands generally exhibit gentler slopes, lower elevations and greater runoff accumulation potential, creating more favorable conditions for infiltration, water storage and groundwater recharge. The persistence of these broad spatial patterns across the evaluated models indicates that the principal hydrological and physiographic controls on RWH suitability were consistently represented.

Despite this overall spatial agreement, notable differences occurred in the extent and configuration of suitability classes predicted by individual algorithms. Such variations are expected because the models employ different optimization strategies and represent nonlinear relationships among predictor variables differently. GBM generated relatively localized high-suitability zones, potentially reflecting greater dependence on dominant predictors such as elevation and drainage density. In comparison, XGBoost produced smoother spatial transitions and more extensive favourable zones, reflecting interactions among rainfall, terrain and land-cover characteristics[103]. These differences demonstrate that model selection can influence not only predictive performance but also the spatial configuration and extent of the resulting suitability classes.

Comparative predictive assessment identified XGBoost as the highest-performing model, with an ROC-AUC of 0.78, followed by SVM (0.76), AHP (0.73), CatBoost (0.69) and GBM (0.68). However, differences in numerical AUC values did not necessarily correspond to statistically significant improvements. Pairwise DeLong tests showed no significant difference between XGBoost and SVM ($p=0.530$) or between XGBoost and AHP ($p=0.129$). In contrast, XGBoost significantly outperformed CatBoost ($p=0.015$) and GBM ($p=0.007$) (Supplementary Table S5). Thus, although XGBoost provided the highest overall discriminatory performance, its improvement over SVM and AHP was incremental rather than statistically decisive. This finding indicates that increasing model complexity does not necessarily produce a proportional improvement in predictive discrimination[104]. Nevertheless, ML approaches provide important methodological advantages because they can represent nonlinear interactions, reduce dependence on subjective expert-derived weighting and accommodate complex multidimensional environmental datasets[105].

Variable-importance analysis further revealed differences in the internal behaviour of the evaluated models[106]. GBM relied comparatively strongly on a limited number of predictors, particularly elevation and drainage density, whereas XGBoost distributed predictive importance more broadly among slope, topographic wetness index, geology, rainfall and other environmental variables. This more distributed predictor structure may partly explain the robustness of XGBoost in a heterogeneous semi-arid landscape[107]. Rainfall, elevation and drainage density consistently emerged among the influential controls on RWH suitability[108]. Rainfall determines the amount of water potentially available for harvesting, while elevation influences regional runoff pathways and terrain organization[109]. Drainage density affects the concentration and routing of surface runoff and therefore influences opportunities for water interception, storage and infiltration[110]. Hillshade showed comparatively low predictive importance, suggesting that its contribution was secondary and potentially related to local terrain configuration and associated variations in surface energy and soil-moisture conditions[111].

Confusion-matrix evaluation provided additional evidence of the performance of XGBoost. The model correctly classified 40 of the 45 validation samples, corresponding to an overall accuracy of 88.89% and a Cohen's kappa coefficient of 0.64. It additionally achieved a precision of 92.0%, recall of 88.46%, specificity of 89.47%, F1-score of 90.20% and Matthews correlation coefficient (MCC) of 0.78 (Table 5). Collectively, these metrics indicate good classification performance and substantial agreement between observed and predicted classes. However, the moderate ROC-AUC of 0.78 suggests that the model should primarily be interpreted as a spatial decision-support tool rather than as a definitive predictive system. The absence of statistically significant AUC differences between XGBoost, SVM and AHP further reinforces the need to distinguish numerical improvements from statistically supported improvements in predictive performance[6].

Several limitations should be considered when interpreting these findings. First, the reliability of RWH suitability mapping remains dependent on the quality, spatial resolution and temporal consistency of the input datasets. Uncertainty associated with the digital elevation model, land-use/land-cover classification and climatic datasets can propagate through the modelling framework and influence final suitability estimates[112]. Second, the reference inventory contained a relatively limited number of observations and model validation relied on a conventional random train-test partition. Spatial dependence between geographically adjacent training and validation observations therefore cannot be completely excluded. Such spatial autocorrelation can lead to optimistic estimates of predictive performance when random cross-validation strategies are applied to spatial datasets. Moreover, the

Table 5 Confusion matrix and comprehensive validation statistics of the best-performing XGBoost model using the independent testing dataset.

Category	Metric	Formula	Value
Confusion matrix	True positives (TP)	Suitable correctly classified	23
	False negatives (FN)	Suitable classified as unsuitable	3
	False positives (FP)	Unsuitable classified as suitable	2
	True negatives (TN)	Unsuitable correctly classified	17
Dataset information	Total validation samples (N)	TP + FP + FN + TN	45
	Correctly Classified Samples	TP + TN	40
	Misclassified samples	FP + FN	5
Overall performance	Overall accuracy (%)	(TP + TN)/N × 100	88.89
	Misclassification Rate (%)	(FP + FN)/N × 100	11.11
	Cohen's Kappa coefficient	$k = \frac{p_0 - p_e}{1 - p_e}$ where: P_o = Observed Agreement (Overall Accuracy) P_e = Expected Agreement by Chance	0.64
	Matthews correlation coefficient (MCC)	$\text{MCC} = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$ where: **TP** = True Positives **TN** = True Negatives **FP** = False Positives **FN** = False Negatives	0.78
Class-wise metrics	Precision (Positive predictive value) (%)	TP/(TP + FP) × 100	92.00
	Recall (Sensitivity) (%)	TP/(TP + FN) × 100	88.46
	Specificity (%)	TN/(TN + FP) × 100	89.47
	Negative predictive Value (%)	TN/(TN + FN) × 100	85.00
	F1-Score (%)	2PR/(P + R) × 100	90.20
	Balanced accuracy (%)	(Sensitivity + Specificity)/2	88.97
ROC statistics	Area under curve (AUC)	ROC analysis	0.78
	Standard error of AUC	DeLong method	0.056
	95% Confidence interval of AUC	Lower–upper bound	0.67–0.89
Statistical Significance	Youden's index	Sensitivity + Specificity − 1	0.779
	McNemar's χ^2	Pairwise Classification Test	0.20
	McNemar's p-value	Significance Test	0.65
	95% Confidence Interval of Accuracy (%)	Binomial Exact Method	76.2–96.0
	95% Confidence Interval of Kappa	Bootstrap Method	0.46–0.82

*TP: True Positive; TN: True Negative; FP: False Positive; FN: False Negative; AUC: Area Under the Receiver Operating Characteristic Curve; MCC: Matthews Correlation Coefficient. Validation statistics were computed using the independent testing dataset comprising 45 samples.

absence of an independent external validation dataset restricts assessment of model transferability beyond Jaipur district.

Additional uncertainty arises from predictors such as hillshade and lineament density, whose relationships with hydrological processes may be indirect and dependent on local hydro geomorphological conditions. Although these variables are frequently incorporated into groundwater and watershed assessments, their contribution should be further examined using sensitivity analysis, predictor-ablation experiments and independent validation across contrasting environments. The present framework also focuses primarily on biophysical suitability and does not explicitly incorporate socioeconomic and implementation constraints. Factors such as land ownership, infrastructure accessibility, agricultural practices, construction costs, local water demand and community participation may substantially influence whether technically suitable RWH locations can be implemented and maintained in practice.

Future research should therefore extend the framework beyond improvements in predictive accuracy alone. Hybrid approaches integrating ML with physically based hydrological knowledge could improve both process representation and predictive reliability. Deep-learning architectures, including convolutional and recurrent neural networks, may also be evaluated where sufficiently large spatial and temporal datasets are available. Explainable artificial intelligence approaches such as SHAP could further quantify the direction, magnitude and interactions of predictor effects, improving model transparency and stakeholder interpretation. More importantly, larger and independently developed RWH inventories, spatial block cross-validation, external validation across contrasting hydro geomorphological regions and uncertainty analysis should be incorporated into future assessments. Such developments would provide a stronger basis for determining whether RWH suitability models remain reliable under changing climatic conditions, land-use trajectories and regional environmental settings.

Conclusion

The present study demonstrated the utility of integrating machine learning algorithms and geospatial techniques for delineating rainwater harvesting suitability zones in Jaipur district, Rajasthan, a semi-arid region facing increasing water stress, rapid urbanization, and considerable hydroclimatic variability. By incorporating eleven environmental and topographic parameters within both machine learning and multi-criteria decision-making frameworks, the study generated spatially explicit maps capable of identifying priority areas for RWH interventions. The spatial analysis consistently identified the northern sector of Jaipur as having relatively low suitability for RWH implementation because of steeper terrain and reduced infiltration potential, whereas the central and southern sectors exhibited moderate to high suitability owing to favorable topographic and hydrological conditions. These findings highlight the importance of considering local geomorphological variability when planning water conservation interventions in semi-arid environments.

The developed ML-GIS framework provides a valuable decision-support tool for prioritizing RWH locations and optimizing site-specific water resource management strategies. However, the framework should not yet be regarded as a fully operational predictive system. The inventory database was relatively limited, validation was based on a conventional random train-test partition, and independent external datasets were unavailable to assess model transferability beyond the study area. Furthermore, the use of generalized datasets, such as the FAO/UNESCO Global Soil Map, and the exclusion of socioeconomic factors, including land tenure, infrastructure availability, and community participation, may have introduced additional uncertainties into the suitability assessment.

Consequently, future investigations should focus on developing larger and more representative inventory datasets, implementing spatially explicit and cross-regional validation frameworks, incorporating higher-resolution environmental datasets, and integrating socioeconomic and climatic projections into the modelling process. The application of explainable artificial intelligence techniques and hybrid deep learning architectures may further enhance model interpretability and predictive performance. Such advancements would improve the robustness, transferability, and practical applicability of RWH suitability models and support sustainable water resource management in semi-arid regions facing increasing climatic and anthropogenic pressures.

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Singh): Conceptualization, Methodology, Supervision, Writing – review & editing. A.S. (Abhishek Sharma): Methodology, Validation, Writing – review & editing. A.S. (Amenjor Senagah): Supervision, Writing – review & editing, Project administration. A.K.T. (Aditya Kumar Tiwary): Data curation, Visualization, Investigation. C.T. (Chandni Thakur): Formal analysis, Validation, Writing – review & editing. M.A.M. (Mohamed A. Mattar): Conceptualization, Supervision, Funding acquisition, Writing – review & editing.

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Data availability All the data utilized in this work is openly added in the manuscript.

Declarations

Competing interests The authors declare no competing interests.

Ethics approval This study did not involve human participants, human-derived data, or animals. Therefore, ethical approval and informed consent were not applicable.

The use of generative AI Generative artificial intelligence (AI) tools were used during the preparation of this manuscript to assist with language refinement, improvement of readability and presentation, and the development and enhancement of selected schematic diagrams and graphical elements. The AI-assisted outputs were critically reviewed, verified, and revised by the authors to ensure scientific accuracy, consistency, and relevance. Generative AI was not used to generate, fabricate, or manipulate research data, statistical results, or scientific findings. The authors take full responsibility for the accuracy, originality, interpretation, and final content of the manuscript.

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